

PHYSICS GRADE 10: PRESSURE

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.1 (9 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Physics
Strand	Strand 1.0: Mechanics and Thermal Physics
Sub-Strand	Sub-Strand 1.1: Pressure
Total Duration	9 lessons × 40 minutes = 360 minutes total
Sub-Strand Content	– Atmospheric pressure and its effects – Factors affecting pressure in liquids – Application of the equation $P = \rho gh$ to determine pressure in liquid – Transmission of pressure in fluids – Applications of pressure in fluids (drinking straw, siphon, hydraulic jack, hydraulic brakes, dam walls)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - describe atmospheric pressure and its effects - investigate the factors affecting pressure in liquids - apply the equation $P = \rho gh$ to determine pressure in liquids - explain the transmission of pressure in fluids - describe the applications of pressure in fluids in day-to-day life - appreciate the applications of pressure in fluids in day-to-day life
Core Competencies	– Critical thinking and problem solving: the learner analyses data from investigations on pressure in liquids and applies $P = \rho gh$ to solve problems. – Communication and collaboration: the learner works in groups during experiments and shares findings and explanations with the class. – Creativity and imagination: the learner designs and improvises simple devices such as a hydraulic lift using syringes. – Learning to learn: the learner revises the class model of pressure as new evidence is gathered.
Core Values	– Responsibility: the learner takes different roles and handles apparatus and water safely during investigations. – Unity: the learner works cooperatively with peers in groups to collect and share evidence. – Respect: the learner values diverse ideas when discussing explanations of the phenomena.
Science & Engineering Practices	– Asking questions and defining problems: students generate questions about why a sharp knife cuts better and add them to the Driving Question Board. – Developing and using models: students build and revise a class model of pressure, force and area. – Planning and carrying out investigations: students test how force, area, depth and liquid density affect pressure.

	– Analyzing and interpreting data: students interpret measurements to find relationships between pressure, depth and density. – Using mathematics and computational thinking: students apply $P = F/A$ and $P = \rho gh$. – Constructing explanations and designing solutions: students explain everyday phenomena and design a simple hydraulic device.
Pertinent & Contemporary Issues (PCIs)	– Safety and security: the learner handles knives, glassware and water safely during investigations. – Environmental education: the learner discusses dam construction and the sustainable use of water resources such as Lake Victoria and the Tana River dams.
Career Connections	– Civil engineer: designing dam walls such as Masinga Dam and foundations that withstand pressure. – Mechanic: servicing hydraulic jacks and brakes in garages in Nairobi and Mombasa. – Pilot and aviation technician: dealing with cabin pressure and altitude. – Chef or butcher: choosing and sharpening knives for efficient cutting. – Meteorologist: studying atmospheric pressure to forecast weather at the Kenya Meteorological Department.
Focus for Lessons	This unit uses the puzzle of a sharp versus a blunt knife to build the concept of pressure as force per unit area. Students move from solids to liquids and the atmosphere, using investigations and simple models to explain dams, hydraulic jacks, ears popping and camels on sand. The unit is anchored by a Driving Question Board and a class model revised each lesson.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why does a sharp knife cut better than a blunt knife when using the same force? KICD KEY INQUIRY QUESTIONS: 1. What is pressure? 2. How does area affect pressure? 3. What factors affect pressure in fluids? 4. Why does pressure increase with depth? 5. What is atmospheric pressure? 6. How is pressure transmitted in fluids? 7. How is pressure used in everyday life?
Anchoring Phenomenon	In a Kenyan kitchen, a mama preparing sukuma wiki, tomatoes and nyama choma uses the same hand force with two knives. The sharp knife slices easily through a tomato and meat, but the blunt knife with the same push barely cuts and just squashes the food. The force and the person are the same, yet the results differ greatly. Students wonder what is different between the two knives that makes the same force produce such different effects.
Storyline Thread	Lesson 1: Anchoring phenomenon - students predict why the sharp knife cuts better than the blunt one, observe both cutting, and create the DQB and initial model. Lesson 2: What is pressure? Students investigate force and area using a brick or books on sand/flour and define $P = F/A$. Lesson 3: How does area affect pressure? Students calculate pressure for knives, camel feet versus shoes, and explain the sharp knife. Lesson 4: Pressure in liquids - students investigate pressure with depth using a can with holes and relate it to dam walls. Lesson 5: Factors affecting pressure in liquids - depth, density and direction; students compare water and salt water or oil. Lesson 6: Applying $P = \rho gh$ - students derive and use the equation to calculate pressure in lakes and dams like Lake Victoria. Lesson 7: Atmospheric pressure and its effects - inverted glass with cardboard, crushed can, ears popping at altitude, and drinking straw.

	Lesson 8: Transmission of pressure in fluids - Pascal's principle with syringes, hydraulic jack and brakes. Lesson 9: Applications and consolidation - siphon, dams and hydraulic devices; students finalise the class model and revisit the DQB to explain the knife phenomenon.
Supporting Phenomena	– Dam walls (e.g. Masinga Dam) are thicker at the bottom than at the top - Why? – A small push on a hydraulic jack lifts a heavy car at a garage - How? – Water doesn't fall from an inverted glass covered with cardboard - Why? – Ears pop when travelling to high altitudes (e.g. Nairobi to Mombasa flight or up Mt. Kenya) - Why? – Camels walk easily on sand while people sink - Why?

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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LESSON 1 (40 minutes (one lesson)): Anchoring Phenomenon: Why Does the Sharp Knife Cut but the Blunt Knife Only Squash?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the unit by presenting the knife phenomenon, so that learners notice that the same force can give very different results, record their own ideas, build a Driving Question Board (DQB) and draw an initial model.
Knowledge	Learners observe that Mama Wanjiru pushes with the same force of 40 N on both knives, but only the sharp knife (edge 0.1 mm thick) cuts the tomato and nyama choma, while the blunt knife (edge 1 mm thick) only squashes them. They identify what is the same (force, person, food, knife size) and what is different (thickness of the edge). No formula and no calculation are used in this lesson.
Skills	Making predictions; observing carefully and recording evidence; asking questions; developing and using an initial model (drawing); communicating and collaborating in groups; Science and Engineering Practices: asking questions, developing and using models, planning and carrying out investigations (observation).
Attitudes	Curiosity about everyday kitchen events; respect for the ideas of others (Respect); responsibility in group roles (Responsibility); willingness to say 'I am not sure yet' and to revise ideas with evidence.
Key Inquiry Question	What is pressure? How is pressure used in everyday life? (Launch question: Why does a sharp knife cut better than a blunt knife when using the same force?)
Purpose in Storyline	This is the opening lesson. It gives the class a shared experience of the phenomenon, which every later lesson returns to. Lesson 2 will take up the DQB question 'What is different between the knives if the force is the same?' and the idea that the force is the same. Learners leave with a first model and a DQB that is revisited and revised in Lessons 2 to 9.
Safety Notes	Knife safety: the teacher or one trained adult handles the sharp knife. Learners do not touch the blades. Use a table knife or a vegetable knife with a clean, covered cutting board, cut away from the body, and keep the knives on a tray when not in use. If real knives are not allowed, use the video or the teacher demonstration with a plastic knife and a ruler-edge to show the blunt case. Wash tomatoes and meat/boiled sweet potato pieces; do not let learners eat the demonstration food. Keep the floor dry and the area clear.

B. LESSON OVERVIEW

Learners meet the opening phenomenon: in a Kisumu kitchen, Mama Wanjiru is preparing sukuma wiki, tomatoes and nyama choma. She pushes with the same 40 N force on two knives of the same size and shape. The sharp knife (edge 0.1 mm thick) slices easily through a tomato and meat, while the blunt knife (edge 1 mm thick) barely cuts and just squashes the food. Learners first predict individually and record initial ideas, then observe a demonstration of both knives (with the teacher using a force meter or a described 'same push' approach), and make a list of what is the same and what is different. In the Explain phase they use talk and drawings to propose reasons, without being given the answer. The class builds a Driving Question Board with the main question and learners' own questions, sorts them with the KICD key inquiry questions, and each learner draws an initial model showing why the two knives give different results. The lesson sets the learning goals for the rest of the sub-strand and deliberately introduces no formula or calculation.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: AP Physics 1 review of Forces and Newton's Laws <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Pressure and Force (Flexbook) Source: CK-12  Search ARES for similar readings	Learners sit in groups of four. The teacher tells the story of Mama Wanjiru in her Kisumu kitchen, cutting tomatoes, sukuma wiki and nyama choma for supper, and shows a picture or a drawn scene of two knives. Learners are told: she pushes with the same hand force of 40 N on each knife, with the same person, the same food and two knives of the same size and shape. Each learner silently writes in their notebook: (1) what they think will happen with each knife, and (2) why. They then draw a quick sketch of the two knife edges and the food. Learners share predictions with a partner, then with the group, and one group reporter records the group's best idea on a sticky note or a small paper.	Display the scene and say: "Mama Wanjiru pushes with exactly the same force on both knives. Before I show you anything, what do you think will happen?" Give a silent writing time of 2 minutes. Say: "There are no wrong answers today; I want your honest first ideas." Use WAIT TIME of 10-15 seconds after asking: "What could be different about the two knives?" Cold-call 4 learners from different groups (not only volunteers) and ask each: "Tell us your prediction and one reason." Write every idea on the board under 'Our first ideas' without judging. Ask: "Who has a different idea from what has been said?" and take 2 more responses. Circulate and note common ideas (e.g. 'sharp knife is stronger', 'sharp knife is heavier', 'sharp knife is pushed harder') for later use. Do not tell the correct answer.	Prediction with a written reason (Predict-Observe-Explain, 'P' step). Writing a prediction before seeing evidence makes each learner commit to an idea, so later evidence can be compared with their own thinking and misconceptions become visible and can be tested.	Check notebooks as you circulate: look for (a) a written prediction for both knives, (b) at least one reason given, (c) a sketch of the edges. Tally on a quick class sheet how many learners attribute the difference to force, to knife material, to sharpness, or to something else. Indicator of readiness: at least 3 of 4 groups have a recorded group idea. Note learners who say the force is different, as this is a misconception to revisit in the Observe phase.
Observe Phase  VIDEO: Buoyant force example problems - dubbed(KY) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Pressure and Force Practice Source: CK-12  Search ARES for similar readings	Learners watch a teacher demonstration (or a short video) of Mama Wanjiru's kitchen situation. The teacher cuts a ripe tomato and a piece of cooked meat (or a soft substitute such as a boiled sweet potato) with the sharp knife, then with the blunt knife, using the same pushing force measured on a kitchen scale or a force meter reading 40 N. Learners are given a table with columns: 'What I see for the sharp knife', 'What I see for the blunt knife'. Groups come forward in turns (by group, 20 seconds each) to look closely at the edges using a hand lens and compare how thick each edge looks (sharp about 0.1 mm, blunt about 1 mm). Learners record observations such as: sharp knife slices through the skin and goes into the tomato; blunt knife presses the tomato flat,	Say: "Watch closely and write down only what you can see, not what you think is the reason." Before the demonstration, state the controls clearly: "The push is 40 N each time, the knives are the same size and shape, and the food is the same." Show the force meter or scale reading each time. Pause after the sharp knife: "What did you notice?" WAIT TIME 10-15 seconds, then cold-call 3 learners. Repeat for the blunt knife with 3 more cold-calls. Ask: "Was the push the same? How do you know?" Pass the hand lens round and ask: "What do you notice about the two edges?" Draw the Same/Different chart on the board and fill it with learners' words. Ask one group to explain the chart to the class and have 2 others agree or disagree: "Do you agree? What	Structured observation with a Same/Different (controlled variables) chart. Separating what stays the same from what changes helps learners to focus on the variable that matters and to rule out force as the difference, which builds the evidence base for later explanations.	Look for learners who record observations (not explanations) in the table, and who correctly list force, person, food and size as the same and the edge thickness as different. Indicator: at least 80% of learners write 'the force is the same' in the chart. Ask 3 learners individually: "What is the same, and what is different between the two knives?" Learners who still say that the force is different should be asked to point to the force meter reading (40 N) as evidence.

	juice squeezes out, and the skin does not break. They then complete a 'Same / Different' chart: Same - force of 40 N, same person, same food, same size and shape of knife, same length of 5 cm touching the food. Different - thickness of the edge (0.1 mm and 1 mm) and the result (cut or squashed).	would you add?" Remind: "Keep the knives on the tray; only I handle the blades." Do not introduce the word pressure or any formula yet.		
Explain Phase  VIDEO: More on Normal force (shoe on floor) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Pressure and Force (Flexbook) Source: CK-12  Search ARES for similar readings	Using the evidence from the Same/Different chart, each group discusses the puzzle: "The force is the same, the person is the same, yet one knife cuts and the other squashes. What could be causing this?" Groups propose at least two possible reasons and write them on a poster, with a drawing of each knife edge touching the tomato (for example, a very thin edge and a thicker edge pressing on the skin) and arrows showing the push. A 'gallery talk' follows: one member stays to explain while others visit other groups, and then they swap. Learners link their ideas to other experiences: why a panga or jembe is sharpened by farmers in Kericho and the Rift Valley, why a needle goes into cloth easily but a blunt nail does not, why a tailor's pin is pointed, and why walking on tiptoes with thin heels is different from flat shoes in soft mud. They finish by writing one sentence beginning: "I think the sharp knife cuts better because ... but I am still wondering ...".	Say: "Talk with your group. Use the evidence from our chart to support your idea." Ask: "If the force is the same, what else could be making the difference?" WAIT TIME 10-15 seconds before taking answers. Cold-call 4 groups to share. Prompt with: "What does that tell us about the edge?" and "What evidence did you see for that?" Do not confirm or correct ideas; respond with "Say more about that" and "Does anyone see it differently?" Ask learners to give a real-life example (panga, jembe, needle, nail, matatu tyre in mud) and ask: "How is this like Mama Wanjiru's knives?" During the gallery talk, listen and note which groups mention the edge, the area touching the food, or the 'sharpness' only. Ask groups with only 'it is sharper' to push further: "What does 'sharper' mean in terms of what touches the food?" Keep the idea of area as a hint, not an answer, and do not give the formula; tell learners that Lessons 2 and 3 will help them test their ideas.	Claim-Evidence-Reasoning talk with a gallery walk. Learners state an idea (claim), point to what they saw (evidence) and give a reason, which trains them to use observation to support explanations rather than guess, while hearing other groups' ideas widens thinking.	Listen for ideas that move toward the thin edge touching the food at a small place or area (even if not the word area). Indicators: each group's poster has at least one claim linked to an observation; at least 2 groups mention the edge thickness or the small part touching the food. Use a thumb check: "Show me thumbs up if you can say what stayed the same." Note misconceptions (e.g. 'the sharp knife is pushed harder', 'it is made of a stronger metal') for Lessons 2 and 3.
Driving Question Board (DQB) Creation  VIDEO: More on	The teacher posts the driving question on a large chart in the centre of the classroom wall: 'Why does a sharp knife cut better than a blunt knife when using the same force?' Individually, learners write	Say: "What questions do you have now that you want to be able to answer by the end of this unit?" Give 3 minutes of silent writing and WAIT TIME of 10-15 seconds after the first group shares. Cold-call 5	Driving Question Board (question clustering). Writing, sorting and linking learners' own questions to the unit's key inquiry questions gives learners ownership of the learning path and shows how each	Check that every learner has posted at least 2 questions and that at least 3 clusters appear on the board. Indicators: questions mention the edge, area, force or something related; at least 2

Newton's third law <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Pressure and Force Practice Source: CK-12 Search ARES for similar readings	2 questions on sticky notes that they want answered (for example: 'Is it the edge that makes it cut?', 'Does the size of the part touching the food matter?', 'Why does a camel not sink into sand?', 'Why do dams have thick walls at the bottom?', 'Why do my ears pop in Nairobi?'). In groups, they sort their questions into clusters, such as 'About the knives and the edge', 'About force and area', 'About liquids and water', 'About air' and 'About machines like the jack'. Each group places one cluster on the DQB and explains it in 20 seconds. The teacher then shows the 7 KICD key inquiry questions (What is pressure? How does area affect pressure? What factors affect pressure in fluids? Why does pressure increase with depth? What is atmospheric pressure? How is pressure transmitted in fluids? How is pressure used in everyday life?) and learners match their own questions to them or add any that are missing.	learners to read a question aloud and to say where it should go on the board: "Where does your question belong, and why?" Group similar questions and ask: "What do these questions have in common?" Highlight questions that connect to Kenyan contexts (camels in Turkana, Masinga Dam, Lake Victoria, Mzee Otieno's jack). Praise good questions, not correct answers: "That is a question we can investigate." Show the 7 KICD questions and ask 2 learners to match one of their notes to a KICD question. Tell learners that the board will stay up and be used in every lesson, and that questions will be moved to 'answered' as the class gathers evidence. Do not answer any question today.	later lesson answers part of the big question.	questions go beyond the knife (such as camels, dams, air). Use a quick exit tally: count how many learners wrote a question about 'area' or 'edge'. Watch for learners who only write 'why' questions without a clear subject and coach them to make the question more specific.
Model Building Phase VIDEO: Gravitational forces <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Gas Pressure and Force (Flexbook)	Each learner draws an initial model on a half-page of paper (or in the notebook) to explain the phenomenon: two diagrams side by side showing the sharp knife and the blunt knife pressing on a tomato, with (a) arrows for the force of the hand, (b) the shape and thickness of each edge, (c) the part of the edge touching the food, and (d) labels of what happens (cuts or squashes). Learners add a short caption: 'My idea is that ...' and a box marked 'What I am not sure about'. In pairs, they swap models, give one 'star' (something clear) and one 'wish' (something to add). A few	Say: "Draw what you think is happening where the knife touches the tomato. Show the force and show the edge." Give 6 minutes of drawing and circulate, asking: "Where is the force in your drawing?" and "What is different about the two edges in your picture?" WAIT TIME 10-15 seconds after each prompt. Cold-call 3 pairs to give a star and a wish after the swap. Select 3 models that show different ideas (for example force different, edge thickness, small touching area) and display them. Ask: "What do these models agree on? Where do they differ?" Write the class's first	Initial model development and peer review (star and wish). Drawing a model makes thinking visible and shows gaps. Peer feedback and a shared class model give a baseline against which later evidence in Lessons 2 to 9 can be used to revise and improve explanations.	Collect or photograph the models. Indicators of a developing model: the force arrow is shown the same on both knives; the sharp knife's edge is drawn thinner or touching a smaller place; the result (cuts / squashes) is labelled. Success criterion: at least 70% of learners show the same force on both knives and a difference in the edge. Learners with force drawn bigger for the sharp knife are flagged for follow-up in Lesson 2. Check that each model has a 'not sure about' box filled in.

<i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	volunteers (chosen by the teacher so different ideas are heard) show models on the board. The class agrees on a shared first model statement written on a chart paper, in learners' own words, for example: 'The knives get the same push, but the sharp one touches the food in a smaller place, so it cuts.' The teacher marks this as 'Version 1 - to be tested' and records it beside the DQB. Finally, learners complete the 'What we observed / What we learned / How we explain' table with the teacher.	model statement on chart paper in learners' words and label it 'Version 1'. Say: "This is our starting model. In the next lessons we will test it, and we will change it when evidence tells us to." Link to next lesson: "Next time we look at a brick on flour to find out what changes when only one thing changes." Tell learners to keep the model, because they will revise it in later lessons. Do not give formulas or numerical calculations.		
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D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners accept that the force was the same for both knives, and which evidence (the force meter reading, the Same/Different chart) convinced them? (2) Which ideas did learners offer for the difference (force, material, sharpness, edge thickness, small area), and how many groups moved toward the edge or the small touching place? (3) Did I avoid giving the answer or the word 'pressure' and the formula, and where was I tempted to? (4) Were all learners, including quiet ones, cold-called at least once, and did my WAIT TIME of 10-15 seconds produce richer answers? (5) Were the DQB questions varied and linked to Kenyan contexts (camels in Turkana, Masinga Dam, Lake Victoria, Mzee Otieno's jack, ears popping in Nairobi)? (6) Was knife safety observed throughout? (7) What do I need to adjust for Lesson 2, where learners test what changes when only the area changes, using the brick on flour or sand? Record names of learners who thought the force was different, and plan a short check with them next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Mama Wanjiru pushes with the same force of 40 N on a sharp knife (edge 0.1 mm thick) and a blunt knife (edge 1 mm thick). The sharp knife slices easily through the tomato and nyama choma; the blunt knife barely cuts and only squashes the food. The person, the force, the food and the size and shape of the knives are the same; the thickness of the edge and the result are different.
What did I learn?	The same force can give very different results. Something other than the force must be different between the two knives. Our first ideas point to the edge and the small place where the knife touches the food. We have not yet explained it with evidence, and we have many questions on our Driving Question Board.
How does this explain the phenomenon?	Our first model (Version 1, to be tested): the knives get the same push, but the sharp knife has a thinner edge and touches the food in a smaller place, so it cuts, while the blunt knife touches in a bigger place and only squashes the food. We will test and improve this model in the next lessons using evidence.

LESSON 2 (40 minutes): What is Pressure? Force and Area on Flour

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate how the same weight of a brick makes dents of different depths in flour or sand when different faces rest on it, and use the results to define pressure as force per unit area ($P = F/A$) with the unit pascal (Pa).
Knowledge	Pressure is the force acting normally (at right angles) per unit area on a surface: $P = F / A$. Pressure depends on both force and area. The unit is the pascal (Pa), where $1 \text{ Pa} = 1 \text{ N/m}^2$. Weight $W = mg$ with $g = 10 \text{ N/kg}$. The 2 kg brick has weight 20 N. Large face 0.02 m^2 gives 1,000 Pa, medium face 0.01 m^2 gives 2,000 Pa, small face 0.005 m^2 gives 4,000 Pa.
Skills	Planning and carrying out investigations (controlling variables: same brick, same weight, same flour); analysing and interpreting data (comparing dent depth with area); using mathematics (converting cm^2 to m^2 and calculating F/A); constructing explanations (defining pressure from evidence).
Attitudes	Curiosity about everyday events in the kitchen, honesty in recording what is seen, and teamwork and respect when sharing results with peers.
Key Inquiry Question	What is pressure?
Purpose in Storyline	In Lesson 1 learners saw that Mama Wanjiru pushes with the same force on a sharp and a blunt knife but gets different results, and they put the question about what differs between the knives on the DQB. Here learners find, using a brick on flour, that the same force on a smaller area makes a deeper dent, and they name this quantity pressure with $P = F/A$. They link back to the knives only qualitatively (same force, different area). The knife calculations come in Lesson 3.
Safety Notes	Place a tray under the flour and keep it on the floor or a low desk. Bricks are heavy (2 kg): lift with two hands, lower gently and keep feet clear. Avoid dropping the brick. Wipe up spilt flour or sand at once so no one slips. Do not blow flour dust; wash hands after the activity.

B. LESSON OVERVIEW

Learners start with Mama Wanjiru's two knives in Kisumu: the same force of 40 N, yet different results. They predict what happens when the same 2 kg brick (weight 20 N) is placed on flour on its large, medium and small face. They then test it, measure dent depth and calculate F/A for each face: 1,000 Pa, 2,000 Pa and 4,000 Pa. They explain why the smallest face sinks deepest, define pressure, and add questions to the DQB. They revise the class model by adding the new idea that the force acting on each unit of area (pressure) is what matters. The sharp knife's tiny edge area is linked to this idea qualitatively: same force, different area.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Pressure and Pascal's principle (part 2)	Learners recall Mama Wanjiru's knives from Lesson 1: the same force, different results. The teacher shows the 2 kg brick (20 cm x 10 cm x 5 cm) and a tray of flour.	Say: 'Remember Mama Wanjiru in Kisumu. She pushed with the same force on both knives, but the results were different. Today we use a brick to find out what really	Prediction with reasoning (Think-Pair-Share): learners commit to an idea and a reason before seeing data, which exposes preconceptions such as 'a bigger	Check written predictions for: (1) a clear choice of which face sinks deepest, (2) a reason mentioning area, weight or force, (3) an attempt to link the brick to the knife

<i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Pressure and Force (Flexbook) Source: CK-12 Search ARES for similar readings	Individually, then in pairs, learners draw and write a prediction: the brick is placed gently on the flour first on its large face, then the medium face, then the small face. Will the dents be the same or different? Which face will sink deepest? Why? They connect it to the knives: 'Which knife edge is like the small face of the brick?'	changes.' Hold up the brick and say: 'This brick weighs the same however I place it. Predict what happens on the flour.' Give 2 minutes of silent individual writing. Then WAIT TIME of 10-15 seconds after asking: 'Which face will make the deepest dent, and what is your reason?' Cold-call 4 learners from different groups, including at least 2 who have not yet spoken. Record predictions on the board in three columns (large deepest / small deepest / all the same) without saying which is correct.	face means more force' and makes the later evidence meaningful.	edge. Tally on the board how many learners chose each column. Note learners who think the force changes with the face.
Observe Phase  VIDEO: Pressure and Pascal's principle (part 1) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Pressure and Force Practice Source: CK-12 Search ARES for similar readings	In groups of 4 (roles: brick handler, measurer, recorder, reporter), learners level flour in a tray to a depth of about 3 cm. They place the brick gently on the flour on the large face (20 cm x 10 cm), lift it straight up, and measure the dent depth with a ruler or stick. They smooth the flour and repeat for the medium face (20 cm x 5 cm) and the small face (10 cm x 5 cm). They record in a table: face, length x width in cm, area in cm², dent depth in mm. Then they convert area to m² (200 cm² = 0.02 m², 100 cm² = 0.01 m², 50 cm² = 0.005 m²), and the weight $W = mg = 2 \times 10 = 20 \text{ N}$, and calculate $P = F/A$ for each face: 1,000 Pa, 2,000 Pa, 4,000 Pa.	Model once how to lower the brick without pushing and how to measure depth at the deepest point. Circulate and ask: 'Is the weight of the brick the same for each face? How do you know?' Say: 'Keep the brick, the flour and how gently you place it the same. What is the only thing you change?' After groups finish measuring, WAIT TIME of 10-15 seconds, then ask: 'Which face sank deepest? Which face has the smallest area?' Cold-call 3 groups to give their depths. Guide the calculation on the board: 'Force is 20 N. Area of the large face is 0.02 m². 20 divided by 0.02 is how many newtons per square metre?' Introduce the name pascal: 1 Pa = 1 N/m².	Controlled investigation with data table: holding the brick's weight constant and changing only the area isolates the effect of area. Tabulating area, depth and F/A lets learners see the pattern that the smaller the area, the deeper the dent and the greater the pressure.	Look for: the same brick and gentle placing used for all trials; depth measured at the deepest point; areas correctly converted (cm² to m²); calculated pressures of 1,000, 2,000 and 4,000 Pa; and the order of dent depths matching the order of pressures. Note groups whose dent depths are not in the right order and ask them to repeat the test.
Explain Phase  VIDEO: Pressure at a depth in a fluid <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)	Learners use their table to answer: What stayed the same? What changed? What happened to the dent depth when the area got smaller? They write a claim-evidence-reasoning statement, for example: 'The small face sinks deepest because the same 20 N is spread over only 0.005 m², so	Say: 'Talk with your partner first: why did the small face sink deepest when the weight was always 20 N?' Give 1 minute for pair talk. WAIT TIME of 10-15 seconds, then cold-call 4 learners. Press for evidence: 'Which numbers in your table show that?' Introduce the term pressure as	Claim-Evidence-Reasoning with a word-and-symbol definition: learners tie their claim to numbers from the data, then generalise it into the formula $P = F/A$, so the definition comes from evidence and not from the teacher.	Check that explanations state that the force (weight) is the same, that the area is different, and that pressure is force per unit area. Look for correct units (Pa, N, m²) and for the right answer of 60 Pa in the quick calculation. Common errors to listen for: 'the small face is heavier', 'pressure is the same

 Search ARES for similar videos  HTML: Pressure and Force (Flexbook) Source: CK-12  Search ARES for similar readings	each square metre carries 4,000 N.' They then write their own definition of pressure in words and as $P = F/A$, state the unit (pascal, Pa; 1 Pa = 1 N/m²; 1 kPa = 1,000 Pa) and explain, in a sentence, how the brick on its small face is like the thin edge of the sharp knife: same force, different area.	force acting at right angles per unit area. Write $P = F / A$ on the board with the unit Pa. Check understanding with one quick calculation: 'A 30 N force acts on 0.5 m². What is the pressure?' (60 Pa.) Give 3 learners the chance to explain. Correct the misconception that pressure and force are the same: 'The brick's force is 20 N on every face, but the pressure is not the same.' Use the name 'pressure' and no other term. Do not yet calculate for the knives; say: 'In Lesson 3 we will use this on Mama Wanjiru's knives.'		as force', and using cm² in the formula without converting.
Driving Question Board (DQB) Creation  VIDEO: Pressure at a depth in a fluid - dubbed(KY) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Pressure and Force Practice Source: CK-12  Search ARES for similar readings	Learners return to the DQB from Lesson 1: 'Why does a sharp knife cut better than a blunt knife when using the same force?' Each group writes on sticky notes (a) one thing the brick experiment helped answer (for example, 'the same force on a smaller area gives a deeper dent'), and (b) one new question (for example, 'How much pressure does a knife edge make?', 'Why can a camel walk on sand?', 'Does water press too?'). They place notes under the key inquiry 'What is pressure?' and next to related questions.	Say: 'Look at our board from Lesson 1. Which question can we now partly answer with the brick?' WAIT TIME of 10-15 seconds. Ask 3 groups to read out and place a note. Then say: 'What new question has the brick raised about the knives?' Cold-call 3 learners. Group similar questions, and mark questions about knife numbers and sinking on sand for Lesson 3, and water and depth for later lessons. Tell learners: 'We have a rule $P = F/A$, but we have not yet put the knife numbers in.'	Driving Question Board: sorting new questions beside earlier ones shows learners how today's evidence links to the main phenomenon and gives them ownership of the next steps in the unit.	Check that each group has at least one note that answers or partly answers the DQB using evidence, and one new question. Look for questions that include area, force or pressure. Note any learners whose questions still ask only about 'sharpness' rather than area.
Model Building Phase  VIDEO: Pressure and Pascal's principle (part 2) <i>Related topic: not</i>	Learners open the initial model they drew in Lesson 1 (the knife, the hand and the food) and revise it using today's evidence. In a new colour they: draw the thin edge and the thick edge touching the food, label 'same force', mark the contact area as small or large, and	Say: 'Take your Lesson 1 model. Using a different colour, add what the brick taught you about the knives.' Circulate and ask: 'Where on your model is the area? Where is the force? Which knife do you think has the greater pressure and why?' After 5 minutes, WAIT TIME	Cumulative model revision: adding new ideas in a different colour to the earlier model lets learners see how their thinking has grown and gives a visible record of the link between evidence (brick) and the phenomenon (knives).	Look for models that show the same force on both knives, a smaller contact area for the sharp edge, the formula $P = F/A$, and the brick evidence. Note models that still show the sharp knife with a bigger force, and revisit the 20 N result for the brick with those

<i>an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Gas Pressure and Force (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	add 'pressure = force / area ($P = F/A$)'. They add a small sketch of the brick on its large and small faces with the values 1,000 Pa and 4,000 Pa as supporting evidence. At the bottom they write: 'What I still need to find out: how big is the pressure for each knife?'	of 10-15 seconds, then cold-call 3 learners to explain one change they made. Do not give the knife numbers. Close with: 'Tomorrow we use $P = F/A$ to calculate the pressure under each of Mama Wanjiru's knives.'	learners.
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D. TEACHER REFLECTION

Did learners hold the brick's weight constant and see that only the area changed? Which learners confused force with pressure, and what evidence from the table helped them? Did every group convert cm^2 to m^2 correctly and obtain 1,000, 2,000 and 4,000 Pa? Which learners spoke for the first time through cold-calling? How well did learners link the small brick face to the thin edge of the sharp knife, using the words 'same force, different area'? What will I adjust in Lesson 3 when we calculate 8 MPa and 0.8 MPa for the knives?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	The same 2 kg brick (weight 20 N) made a deeper dent in the flour when it rested on its small face (10 cm x 5 cm) than on its large face (20 cm x 10 cm).
What did I learn?	Pressure depends on both force and area. Pressure is force acting at right angles per unit area: $P = F/A$, measured in pascals (Pa), where 1 Pa = 1 N/ m^2 . The large, medium and small faces gave 1,000 Pa, 2,000 Pa and 4,000 Pa.
How does this explain the phenomenon?	The weight (20 N) was the same on every face, but on the smallest area (0.005 m^2) the force was concentrated, giving the greatest pressure (4,000 Pa) and the deepest dent. This suggests that Mama Wanjiru's sharp knife, with the same force on a much smaller edge area, can give a greater pressure. We will calculate this in Lesson 3.

LESSON 3 (40 minutes (1 lesson)): How Does Area Affect Pressure? Calculating Pressure for Knives, Camel Feet and Shoes

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use $P = F/A$ to calculate and compare pressures when the area changes, and apply the relationship to explain why Mama Wanjiru's sharp knife cuts while the blunt knife only squashes the food under the same 40 N force.
Knowledge	For a constant force, pressure is inversely related to contact area: a smaller area gives a larger pressure and a larger area gives a smaller pressure. Sharp knife: $A = 5 \times 10^{-6} \text{ m}^2$, $P = 8 \text{ MPa}$. Blunt knife: $A = 5 \times 10^{-5} \text{ m}^2$, $P = 0.8 \text{ MPa}$. The sharp edge has 10 times less area and gives 10 times more pressure. A tomato skin or nyama choma surface gives way at about 2 MPa, so 8 MPa cuts and 0.8 MPa only squashes. Camel: 4,000 N over 0.2 m^2 gives 20 kPa. A person: 600 N over 0.04 m^2 gives 15 kPa.
Skills	Calculating pressure with $P = F/A$; converting cm^2 and mm to m^2 and m ; converting between Pa, kPa and MPa; comparing quantities using ratios; analysing and interpreting data (NGSS SEP: Analyzing and Interpreting Data); using mathematical and computational thinking; constructing explanations from evidence; revising a model.
Attitudes	Curiosity about everyday tools and animals; appreciating that physics explains how Kenyan kitchens and desert animals work; precision and honesty in recording calculations; respect for peers' reasoning and collaborative teamwork.
Key Inquiry Question	How does area affect pressure?
Purpose in Storyline	Lesson 1 showed that the same force gives different cutting results and recorded initial ideas and a model with no formula. Lesson 2 taught that pressure depends on both force and area, $P = F/A$ in pascals, using the brick on flour (1,000, 2,000, 4,000 Pa). In Lesson 3 learners use that formula on the actual knife data and on the camel and person, which turns the qualitative idea into numbers (8 MPa vs 0.8 MPa) and gives the first quantitative answer to the driving question. Lesson 4 then moves to pressure in liquids and depth.
Safety Notes	The knives in this lesson are used only as numerical data and for teacher demonstration. If a real knife is shown, the teacher handles it alone, keeps it away from learners and does not use it for cutting in class. Use a blunt table knife or a picture for display. Keep walkways clear when learners move around for the gallery walk.

B. LESSON OVERVIEW

Learners begin by predicting how the pressure under the sharp and blunt knives compares when Mama Wanjiru pushes with 40 N, and how a heavy camel compares with a 60 kg student on sand. They then calculate $P = F/A$ for both knives, convert units and compare the results with the 2 MPa value at which tomato skin or nyama choma gives way. Next they calculate the pressures for the camel and the person and explain why the 6.7 times heavier camel does not sink much more. In the Explain Phase they write a claim-evidence-reasoning explanation of the sharp knife, stating 'same force, different area'. The class adds to the Driving Question Board (DQB) and revises its initial model from Lesson 1 by adding the numbers and the area-pressure relationship.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	(7 min) Learners look at a picture of Mama Wanjiru in her Kisumu	Display the picture and say: "Same hand, same 40 N. Last lesson,	Predict-Commit-Revisit: Learners commit to a prediction and a	Observe whether learners (1) recall that a smaller area gave a

 VIDEO: Pressure and Pascal's principle (part 2) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Pressure and Force (Flexbook) Source: CK-12  Search ARES for similar readings	kitchen with the sharp and blunt knives, both pushed with 40 N. In their notebooks they first recall from Lesson 2 the rule $P = F/A$ and the brick results (1,000, 2,000, 4,000 Pa). They then predict individually: (a) which knife gives the larger pressure on the tomato and by about how many times, given that the edge is 0.1 mm thick versus 1 mm; (b) who presses harder on desert sand in Turkana, a 400 kg camel or a 60 kg student in school shoes, and whether the camel sinks about 6 times deeper. They write a prediction and a reason in one sentence and share with a partner.	with the brick, what did a smaller face do to the pressure? Write it down." Pose: "Predict how many times bigger the pressure is under the sharp edge. Write a number or a guess, with a reason." WAIT TIME 10-15 seconds in silence before anyone speaks. Then say: "The camel is about 6 times heavier than the student. Will it sink about 6 times deeper? Commit to yes or no." Ask for a show of hands, then cold-call 4 learners to give their reasoning without saying whether they are right. Record the predictions on the board as 'Our predictions' to revisit at the end.	reason before calculating. This surfaces the misconception that more weight always means more pressure (ignoring area), and makes the later calculation a test of their own ideas, which builds ownership of the new understanding.	larger pressure in the brick results; (2) give a reason that mentions area or thickness rather than only 'sharp is stronger'; (3) write a number or ratio in their prediction. Note on a class tally how many predict that the camel sinks about 6 times deeper, as an indicator of the weight-only misconception.
Observe Phase  VIDEO: Pressure and Pascal's principle (part 1) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Pressure and Force Practice Source: CK-12  Search ARES for similar readings	(13 min) In groups of 4, learners work from a data card. Task 1, sharp knife: $F = 40\text{ N}$, contact length $5\text{ cm} = 0.05\text{ m}$, edge thickness $0.1\text{ mm} = 0.0001\text{ m}$. They calculate $A = 0.05 \times 0.0001 = 0.000005\text{ m}^2$ ($5 \times 10^{-6}\text{ m}^2$), then $P = 40 / 0.000005 = 8,000,000\text{ Pa} = 8\text{ MPa}$. Task 2, blunt knife: same $F = 40\text{ N}$, edge thickness $1\text{ mm} = 0.001\text{ m}$, $A = 0.05 \times 0.001 = 0.00005\text{ m}^2$ ($5 \times 10^{-5}\text{ m}^2$), $P = 40 / 0.00005 = 800,000\text{ Pa} = 0.8\text{ MPa}$. They compare: the area is 10 times smaller and the pressure 10 times larger. They compare both values with the 2 MPa at which a tomato skin or nyama choma surface gives way: 8 MPa is greater than 2 MPa so it cuts, 0.8 MPa is less than 2 MPa so it only squashes. Task 3, camel and person: camel weight $400 \times 10 = 4,000\text{ N}$, four feet of 0.05 m^2 each $= 0.2\text{ m}^2$, $P = 4,000 / 0.2 = 20,000\text{ Pa} = 20\text{ kPa}$. Person: $60 \times 10 = 600\text{ N}$, two shoes total 0.04 m^2, $P = 600 / 0.04 = 15,000\text{ Pa} = 15\text{ kPa}$. Extension: the same camel on	Circulate and say: "Show me the unit conversion first: how many metres is 0.1 mm?" Prompt: "Which number is the same for both knives? Which one changed?" For groups stuck on the powers of 10, say: "Write 0.000005 as 5×10^{-6} and tell me how many times smaller it is than 5×10^{-5}." Before groups move to the camel, stop the class and cold-call 3 groups to give their sharp and blunt pressures. Say: "Compare 8 MPa with 2 MPa. What happens to the tomato? Now 0.8 MPa. What happens?" WAIT TIME 10-15 seconds after each question. Remind groups that $1\text{ MPa} = 1,000,000\text{ Pa}$ and $1\text{ kPa} = 1,000\text{ Pa}$, and to always write units. After the camel results, ask: "Camel 4,000 N, person 600 N, and the pressures are 20 kPa and 15 kPa. Look at your table. What do you notice?" Cold-call 3 learners.	Data table and ratio comparison (Analyzing and Interpreting Data): Learners generate their own data table from $P = F/A$ and compare rows by dividing quantities. Seeing that the area drops 10 times while the pressure rises 10 times for the knives, and that a 6.7 times heavier camel gives only a slightly larger pressure, helps them discover the inverse relationship between area and pressure for a constant force from numbers rather than being told.	Check that learners (1) convert 0.1 mm to 0.0001 m and 1 mm to 0.001 m correctly; (2) get $A = 5 \times 10^{-6}\text{ m}^2$ and $5 \times 10^{-5}\text{ m}^2$; (3) get $P = 8\text{ MPa}$ and 0.8 MPa with correct units; (4) state the 10 times ratio; (5) compare each pressure with 2 MPa; (6) get 20 kPa and 15 kPa. Tally how many groups have all values correct on the first attempt and note common errors (mm to m, wrong powers of ten, forgetting to use weight in N, not multiplying the four feet).

	narrow hooves of total 0.04 m ² would give $4,000 / 0.04 = 100,000$ Pa = 100 kPa. Groups record all results in a table with columns Object, Force (N), Area (m ²), Pressure (Pa).			
Explain Phase  VIDEO: Pressure and Pascal's principle (part 2) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Pressure and Force (Flexbook) Source: CK-12  Search ARES for similar readings	(10 min) Individually, then in pairs, learners write a claim-evidence-reasoning (CER) explanation answering: 'Why does the sharp knife cut the tomato and nyama choma while the blunt knife only squashes it, when Mama Wanjiru uses the same force?' The claim states that the difference is the area, not the force. The evidence uses the numbers: 40 N for both, 5×10^{-6} m ² vs 5×10^{-5} m ² , 8 MPa vs 0.8 MPa, and 2 MPa as the value at which the food gives way. The reasoning uses $P = F/A$: same force, different area, so the smaller area gives the larger pressure. Then pairs answer a second prompt: 'Why does the camel, about 6.7 times heavier than the person, not sink much deeper than the student on the sand in Marsabit?' (its pressure, 20 kPa, is only a little more than 15 kPa because its four wide feet spread the weight over a larger area, 0.2 m ² vs 0.04 m ²). Pairs also say what the narrow hooves value (100 kPa) would mean for the camel.	Say: "Start your answer with the words 'same force, different area'. Then give two numbers as evidence." Show a CER frame on the board: 'Claim / Evidence (numbers with units) / Reasoning ($P = F/A$)'. WAIT TIME 10-15 seconds of silent thinking before pairs talk. Circulate and ask: "Where in your answer is the 2 MPa? Why does it matter?" and "Is the knife cutting because of a bigger force or a bigger pressure?" Cold-call 3 pairs to read their explanation; after each ask another learner: "What would you add or change?" Correct the language: use 'pressure', not 'stress' or 'strength'. Finally ask: "Which two things are the same and which is different between the knives?" and have 2 learners answer in a full sentence.	Claim-Evidence-Reasoning (Constructing Explanations): Learners connect a claim to numerical evidence and a scientific principle. Writing the reasoning with $P = F/A$ forces them to link the formula to the phenomenon, and applying it to the camel tests that the idea transfers to a new Kenyan context.	Look for explanations that (1) say the force is the same (40 N) and the area is different; (2) quote 8 MPa and 0.8 MPa with units; (3) compare them with 2 MPa; (4) use the words 'smaller area gives larger pressure'; (5) for the camel, mention that a larger area spreads the weight so the pressure is not much larger. Use a 3-point check: claim correct, evidence with units, reasoning with $P = F/A$. Note learners who still say the sharp knife 'pushes harder' for follow-up.
Driving Question Board (DQB) Creation  VIDEO: Pressure and Pascal's principle (part 1) <i>Related topic: not an exact match</i>	(5 min) Learners return to the DQB started in Lesson 1. In their groups they write on sticky notes (a) one question that is now answered, such as 'What is different between the two knives?' with the answer 'The area, so the pressure is 10 times greater (8 MPa vs 0.8 MPa)', and (b) one new question that has arisen, for example 'Does pressure	Say: "Go to our board. Find a question from Lesson 1 that we can now answer with numbers. Move it to the 'answered' column and write the evidence." WAIT TIME 10-15 seconds before groups begin. Then: "What new question has this lesson created for you?" Cold-call 3 groups to read a new question aloud, and	Driving Question Board tracking: Moving questions between 'unanswered' and 'answered' shows learners the progress of their own understanding and keeps the purpose of each lesson tied to the original knife phenomenon. New questions create the need for the next lesson.	Check that (1) at least one answered question includes numerical evidence and units; (2) new questions are testable and linked to pressure, area, force or fluids; (3) all groups contribute. Note the number of questions on liquids and depth to guide Lesson 4.

for this lesson. Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Pressure and Force Practice Source: CK-12 Search ARES for similar readings	also act inside water, like in Lake Victoria?', 'Why do dam walls look thick at the bottom?' or 'What happens if the area is made even smaller?'. They place the notes on the board under the key inquiry question 'How does area affect pressure?'.	group related questions (for example, those about liquids and depth) with a marker, telling learners that Lesson 4 begins to look at them. Do not answer the new questions now.		
Model Building Phase VIDEO: Pressure and Pascal's principle (part 2) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Gas Pressure and Force (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	(5 min) Learners take out the initial model they drew in Lesson 1 (Mama Wanjiru's hand, a knife and food, with arrows for force). Individually they revise it by (a) drawing a close-up of the sharp edge and the blunt edge touching the tomato, with the same 40 N arrow on both; (b) labelling the contact areas $5 \times 10^{-6} \text{ m}^2$ and $5 \times 10^{-5} \text{ m}^2$; (c) adding the pressures 8 MPa and 0.8 MPa and the 2 MPa line for the food giving way; (d) writing the rule 'same force, smaller area gives larger pressure, $P = F/A$'. They add a small side sketch of the camel's wide feet vs the shoes. They then compare with a partner and add one remaining question about what is still unexplained.	Say: "Add to your model. What must you draw to show that the force is the same but the pressure is different?" WAIT TIME 10-15 seconds, then circulate asking: "Where is the area on your drawing? Where is the 2 MPa?" Cold-call 2 learners to show their drawing and explain it in one sentence using 'same force, different area'. Return to the predictions on the board and ask 2 learners to say what was right and what changed. Close with: "Next lesson we ask whether liquids like the water in Lake Victoria also press, and how deep matters."	Cumulative model revision (Developing and Using Models): Learners update the Lesson 1 model with quantities and a mechanism (area changes the pressure when force is constant). Comparing the old and new model makes visible how their explanation has become more complete and predictive.	Check that revised models (1) show the same 40 N on both knives; (2) label different contact areas; (3) show 8 MPa and 0.8 MPa with units; (4) show the 2 MPa threshold; (5) state the relationship between area and pressure. Use a quick 0-2-4 rubric score and note which models still show the sharp knife with a larger force.

D. TEACHER REFLECTION

After the lesson consider: Which learners could convert mm to m and cm^2 to m^2 without help, and which need more practice before Lesson 6? Did learners recognise that pressure, not force, differs between the knives, or do some still say the sharp knife pushes harder? Did the camel example help them to see that more weight does not always mean much more pressure, or did it confuse them? Which questions on the DQB point to liquids, so that Lesson 4 can begin from learners' own questions? What changes would help in time management, particularly in the Observe Phase calculations?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Same force of 40 N from Mama Wanjiru on both knives, but the contact area differs: sharp knife $5 \times 10^{-6} \text{ m}^2$, blunt knife $5 \times 10^{-5} \text{ m}^2$. Calculated pressures: sharp 8 MPa, blunt 0.8 MPa. Camel (4,000 N, 0.2 m ²) gives 20 kPa, the person (600 N, 0.04 m ²) gives 15 kPa, and the camel on narrow hooves (0.04 m ²) would give 100 kPa.
What did I learn?	Using $P = F/A$, for a constant force the pressure is larger when the area is smaller (inverse relationship). The sharp edge has 10 times less area and gives 10 times more pressure. A heavier object can give a similar pressure if its weight is spread over a larger area.
How does this explain the phenomenon?	The sharp knife cuts because, with the same force of 40 N, its thin edge has a much smaller area, so the pressure (8 MPa) is greater than the 2 MPa at which the tomato or meat gives way. The blunt knife, with the same force and a larger area, gives only 0.8 MPa, which is less than 2 MPa, so the food is just squashed. The difference is the area, not the force.

LESSON 4 (40 minutes): Pressure in Liquids and Depth: The Can with Holes and the Dam Wall

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate how liquids exert pressure and how this pressure changes with depth, using a can with holes and a dam wall, so they can add liquids to their explanation of how pressure depends on force and area.
Knowledge	Liquids exert pressure on the walls of their container, and this pressure increases with depth. Water at 5 cm, 10 cm and 15 cm depth gives 500 Pa, 1,000 Pa and 1,500 Pa (qualitative first, values introduced after the observation). A dam with water 30 m deep has 300 kPa at its base, which is why the wall is built thicker at the bottom.
Skills	Predicting, observing and recording, comparing jet distances, identifying patterns (deeper means greater pressure), constructing explanations using evidence, and developing and revising a model.
Attitudes	Curiosity about everyday water systems, respect for teammates' ideas, responsibility when handling water and sharp tools, and appreciation of how physics supports safe dam design in Kenya.
Key Inquiry Question	What factors affect pressure in fluids? Why does pressure increase with depth?
Purpose in Storyline	Lessons 2 and 3 showed that pressure is force per area on a surface ($P = F/A$). Lesson 4 applies this idea to liquids for the first time. Learners find that liquid pressure increases with depth and that dam walls are thicker at the bottom. This is the first step toward asking what else affects liquid pressure (Lesson 5) and how to calculate it (Lesson 6). It also keeps the driving question alive: pressure, not only force, decides what happens at a surface, whether a knife edge or a dam wall.
Safety Notes	Make the holes ahead of time, or have the teacher make them with a hot nail or a pin. Learners must not use sharp tools. Wipe spilt water at once to prevent slipping. Work over a basin, tray or outdoors. Keep electrical items away from water. Use only clean tap or borehole water.

B. LESSON OVERVIEW

Mama Wanjiru in Kisumu pushes with 40 N on both knives, and the sharp knife cuts because pressure = force / area is larger when area is small (Lessons 2 and 3). Now learners ask a new question about the same idea: does a liquid also push on surfaces, and does that push change with depth? Learners predict how water will squirt from a can with three holes at 5 cm, 10 cm and 15 cm, observe the jets, explain the pattern, and link it to a dam such as Masinga with water 30 m deep, where the base experiences 300 kPa. They add questions to the Driving Question Board and add liquids to the class model, with pressure from water acting on the dam wall just as pressure from the knife edge acts on a tomato.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Pressure at a depth in a fluid	Learners recall Mama Wanjiru's knives: same force of 40 N, different area, different pressure. The teacher shows a plastic bottle	Hold up the bottle and say: "Mama Wanjiru's knife pressed on food. Today, water is going to press on something. Where do you think it	Predict-Share-Compare: Writing and sketching a prediction before seeing evidence makes learners commit to their current ideas, so	Check that each learner has a written prediction and a sketch of three jets. Indicators: (1) uses ideas of force or pressure from

Source: Khan Academy (English - US curriculum) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: pressure liquid depth holes	or tin filled with water and three covered holes at 5 cm, 10 cm and 15 cm from the water surface. Learners individually sketch in their notebooks what they expect when the tape is removed: how far will each jet squirt, and why? They write a one-sentence prediction, share with a partner, then with the class. Learners also predict the dam question: Masinga Dam holds water 30 m deep, so should the wall be the same thickness from top to bottom? Why or why not?	presses, and does it press more in some places than others?" Give learners 2 minutes to draw and write a prediction. Say: "Turn to your partner and explain your sketch in 30 seconds each." Then WAIT TIME of 10-15 seconds before cold-calling 4 learners to share. Ask: "What made you choose that jet order?" Record 3 different predictions on the board without judging. Ask: "Who has a different prediction?" Do not reveal the answer.	the later observation creates real cognitive conflict. Sharing with a partner then the class exposes the range of ideas (for example, that the top hole squirts farthest) that the evidence will test, and connects back to the pressure ideas from Lessons 2 and 3.	Lessons 2 and 3, (2) gives a reason, not just an answer, (3) predicts the dam wall in some way. Note common ideas such as 'top hole is farthest because it is closest to the air' or 'all holes are equal' to revisit in the Explain Phase.
Observe Phase VIDEO: Pressure at a depth in a fluid - dubbed(KY) Source: Khan Academy (Kiswahili) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: pressure liquid depth holes	In groups of 4, learners use a can or bottle with three holes at 5 cm, 10 cm and 15 cm from the water surface, with the holes covered by tape. Roles: pourer, tape remover, measurer, recorder. They fill the container with water, place it at the edge of a table with a basin below (or outdoors), remove all tapes at once, and observe the jets. They measure how far each jet lands horizontally from the can and record in a table: hole depth (5, 10, 15 cm), distance of jet. They repeat once to check results and keep the water level topped up. Then they are given qualitative first findings and the pressure values at each depth: 500 Pa, 1,000 Pa and 1,500 Pa for water. Finally they watch or study a diagram of a dam with water 30 m deep, where the thick wall is at the base. Learners compare the pressure at 10 m depth (100 kPa) with the base at 30 m depth (300 kPa).	Demonstrate once and give each group their kit. Say: "Remove all three pieces of tape at the same time. Watch where the water lands, and measure from the can to where each jet lands." Circulate and ask: "Which hole squirts farthest? Which squirts the least? What do you notice happens as the water level goes down?" Ask a group: "What does a longer jet tell us about the push of the water?" After data collection, WAIT TIME 10-15 seconds, then cold-call 3 groups for their measurements and draw a class table on the board. Then introduce the values: "Scientists have measured the pressure: 500 Pa at 5 cm, 1,000 Pa at 10 cm, 1,500 Pa at 15 cm." Show the dam diagram: "Water at the Masinga Dam is 30 m deep. At 10 m depth the pressure is 100 kPa, at the base it is 300 kPa. Where would you expect the wall to need more strength?" Remind learners to use units with every number.	Structured Data Collection and Pattern Seeking: Learners gather their own measurements, then match their pattern (lower hole, longer jet) to the pressure values provided. Recording in a table makes the relationship between depth and pressure visible, so learners can spot the pattern themselves before the teacher names it. The dam diagram lets them see the same pattern at large scale.	Observe whether groups: (1) remove tapes together and measure from the same point, (2) record three distances in a table with units, (3) state that the lowest hole squirts farthest, (4) use pressure values of 500 Pa, 1,000 Pa and 1,500 Pa correctly in their table. Ask a learner: "How does the jet at 15 cm compare to the jet at 5 cm?" Look for 'farther' or 'stronger'. Check that no learner is just copying another group's numbers.
Explain Phase VIDEO: Pressure at a	Groups use their data to answer: (a) What does the pattern of jet distances tell you about how	Say: "Use your table to complete: Claim, Evidence, Reasoning." Give 5 minutes. Prompt: "Imagine a	Claim-Evidence-Reasoning (CER): Writing a claim supported by their measured data and a reason	Look for CER paragraphs that: (1) state that pressure increases with depth, (2) cite specific numbers

depth in a fluid Source: Khan Academy (English - US curriculum) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: pressure liquid depth holes	pressure changes with depth? (b) Why does water push sideways out of the holes if only gravity pulls it down? (c) Why is the wall of a dam such as Masinga built thicker at the bottom? Learners write a claim-evidence-reasoning paragraph, such as: 'Claim: Pressure in a liquid increases with depth. Evidence: the hole at 15 cm squirted farthest and the pressure was 1,500 Pa while at 5 cm it was 500 Pa. Reasoning: there is more water above a deeper point, so more weight presses, and the water pushes on the can wall (and out through the hole).' They then compare the dam: 100 kPa at 10 m and 300 kPa at 30 m, and explain why the thick base is needed. Finally, they link back to the knife: the dam wall is like the food and the water pressure like the knife pressure, since a pressure above what the surface can hold causes failure.	column of water above each hole. What is different about the column above the lowest hole?" Then say: "Think-pair-share: why does a deeper point feel more push?" WAIT TIME 10-15 seconds before cold-calling 4 learners. Probe: "How is this like Mama Wanjiru's knife? What was pressure then, and what is it now?" Clarify that in liquids the pressure acts on the container walls and pushes outwards through any opening. Do not derive $P = \rho gh$ yet. Say: "We will find how to calculate this in Lesson 6." Ask: "If the dam wall were the same thickness all the way, what could happen at the base?" Correct any misconception that more water in a wider container always means more pressure at the same depth, noting we will test that in Lesson 5.	forces learners to link observations to the idea of more water above means more weight pressing. Connecting the can to the dam then transfers the idea to a real Kenyan structure, and the knife comparison reinforces that pressure is what decides the outcome at a surface.	such as 1,500 Pa at 15 cm versus 500 Pa at 5 cm or 300 kPa versus 100 kPa, (3) give a reason about more water above, (4) connect to the thick base of a dam. Use a quick thumbs-up check: 'Raise your hand if your reasoning explains why, not just what.' Note learners who say 'the water is heavier at the bottom' (partly right) and ask them to refine it.
Driving Question Board (DQB) Creation VIDEO: Pressure at a depth in a fluid - dubbed(KY) Source: Khan Academy (Kiswahili) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms:	Learners return to the Driving Question Board with the knife question at the centre: 'Why does a sharp knife cut better than a blunt knife when using the same force?' Each learner writes one new question or answer on a sticky note based on today's evidence. Examples: 'Does the shape of the dam or can change the pressure?' 'Does salty water give a different pressure from fresh water?' 'How can we calculate pressure at 30 m?' They also place a note that answers an earlier question (for example, 'Pressure is force per area, and in liquids it increases with depth'). Groups cluster notes under the KICD questions: What factors affect pressure in fluids? Why does	Say: "Take a sticky note. Write one question that today's can and dam made you ask, and one thing you can now answer." Give 3 minutes. Then say: "Place your notes on the board near the knife question." WAIT TIME 10-15 seconds, then cold-call 3 learners to read their notes aloud and say where they belong. Ask: "Which question will we answer next lesson?" Point out questions about density and container shape and say: "Lesson 5 will test those." Mark answered questions with a tick and keep unanswered ones visible.	DQB Questioning and Clustering: Posting and sorting questions makes learners track what they know and what they still need to investigate. Linking today's notes to the knife question keeps the whole unit coherent and shows that each lesson answers part of the original puzzle.	Check that each learner posts at least one question and one answer. Indicators: (1) questions are testable (for example, about density or shape), (2) answers cite depth or the dam, (3) learners cluster notes sensibly. Record which key questions remain open. A learner who writes only 'why is water heavy?' should be prompted: 'What could you change in the can to test that?'

pressure liquid depth holes	pressure increase with depth?			
Model Building Phase  VIDEO: Pressure and Pascal's principle (part 1) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: pressure liquid depth holes	Learners revise the class model that began in Lesson 1 and grew in Lessons 2 and 3. In groups, they redraw a model that now includes liquids: (1) the knife edge pressing on a tomato with arrows showing force and area (from earlier lessons); (2) a can of water with three holes, arrows pushing out of the walls, longer arrows at greater depth, and the values 500 Pa, 1,000 Pa and 1,500 Pa; (3) a dam wall with water 30 m deep, thicker at the base, with arrows showing 100 kPa at 10 m and 300 kPa at the base. They add a short caption: 'Pressure is force per area. In a liquid, pressure increases with depth.' Each group shares one change they made to their model.	Say: "Revise your model. Show what is new today. Use arrows to show where the pressure is bigger." Circulate and ask: "Where is the pressure largest in your drawing? How do you show that? How is the dam wall like the tomato?" Say: "Add labels with units." After 8 minutes, WAIT TIME 10-15 seconds and cold-call 3 groups to share one revision. Ask the class: "What is still missing from our model that we need to explain the knife fully?" Accept answers about density, atmospheric air and hydraulics, and note them for later lessons. Do not introduce $P = \rho gh$.	Cumulative Model Revision: Learners add liquid pressure to their existing model instead of starting again. Seeing how arrows and labels change shows how their understanding of pressure has grown across lessons. Identifying gaps prepares them for Lessons 5 to 9.	Check that models: (1) keep the knife idea of force and area, (2) show longer arrows or larger values at greater depth, (3) include the dam wall thicker at the base, (4) use correct units (Pa, kPa). Ask: "What did you change from last lesson?" Look for a clear revision rather than a copy. Note groups who confuse pressure with force and revisit that with $P = F/A$.

D. TEACHER REFLECTION

Which learners predicted that the top hole would squirt farthest, and did the can evidence change their thinking? Were groups able to link more water above to more pressure, or did they only describe the jets? How well did the dam example help them see the same idea at a large scale? Which DQB questions about density, container shape or calculation should be prioritised in Lessons 5 and 6? What changes to group roles or equipment (holes, tape, basins) would improve data quality next time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	The water from the lowest hole (15 cm) squirted farthest, and the water from the highest hole (5 cm) squirted the least. Pressure values: 500 Pa at 5 cm, 1,000 Pa at 10 cm, 1,500 Pa at 15 cm. A dam with water 30 m deep has 300 kPa at the base compared with 100 kPa at 10 m depth.
What did I learn?	Liquids exert pressure on the walls of their container and in all the places they touch, and this pressure increases with depth. More water above a point means more weight pressing there.
How does this explain the	Dam walls are built thicker at the bottom because the pressure there (300 kPa at 30 m) is much larger than near the top, so the

phenomenon?

wall must withstand more. This extends the knife idea: pressure, not just force, decides what a surface can bear. The sharp knife works because the small area gives a high pressure, and the dam base needs strength because depth gives a high pressure.

LESSON 5 (40 minutes): Factors Affecting Pressure in Liquids: Depth, Density and Direction

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate which factors change the pressure in a liquid. They find that liquid pressure depends on depth and density, not on container shape or amount of liquid, and that it acts equally in all directions at the same depth. This gives them more evidence for the idea of pressure behind Mama Wanjiru's knives.
Knowledge	Liquid pressure increases with depth and with density. At 10 cm (0.10 m) the pressure is 1,000 Pa in water (1,000 kg/m ³), 1,030 Pa in salt water (1,030 kg/m ³) and 920 Pa in cooking oil (920 kg/m ³). Container shape and the amount of liquid do not change the pressure at a given depth. At the same depth the pressure is the same in all directions.
Skills	Predicting, observing and comparing the spurt of liquid from holes at the same depth in different liquids; controlling variables (same depth, different liquid); using evidence to argue a claim; interpreting data in a table; revising a model.
Attitudes	Curiosity about why liquids behave as they do; respect for others' ideas during discussion; responsibility when handling equipment and clearing spills; honesty in reporting what was actually observed.
Key Inquiry Question	What factors affect pressure in fluids?
Purpose in Storyline	Lesson 4 showed that pressure in a liquid increases with depth (can with holes at 5, 10 and 15 cm). Lesson 5 asks what else matters: the liquid itself (density), the shape of the container, the amount of liquid, and direction. It prepares for Lesson 6, where students derive $P = \rho g h$. For the knife question, it extends the idea that pressure depends on the physical quantities involved, not on how big or heavy something looks. In the knives, the force was the same and the area was different. In liquids, depth and density are what make the difference.
Safety Notes	Do the activity over a tray or basin to catch spills, and wipe spills at once because oil and water make floors slippery. Make holes in cans or bottles with a nail or hot needle only by the teacher beforehand, and file down sharp edges. Do not taste the salt water or oil. Wash hands after the activity. Dispose of oil in a container, not down the drain.

B. LESSON OVERVIEW

Mama Wanjiru's knives showed that the same 40 N of force can give very different pressures. Today learners ask what makes the pressure in a liquid larger or smaller. Building on Lesson 4, where lower holes in a can squirted farther, groups compare water, salt water and cooking oil at the same depth of 10 cm and predict which liquid squirts farthest. They then look at tall, wide and narrow containers with the same depth, and at holes pointing in different directions at the same depth. They use the data (1,000 Pa, 1,030 Pa and 920 Pa) to conclude that pressure depends on depth and density, not on container shape or amount of liquid, and acts equally in all directions at the same depth. Learners add new questions to the Driving Question Board and revise the class model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners recall Mama Wanjiru's kitchen and the can with holes	Hold up the three cans and say: 'In Lesson 4 we saw that deeper	Predict-Observe-Explain (POE) with a public prediction tally.	Check that each learner has written a prediction and a reason

 VIDEO: Pressure at a depth in a fluid Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Under Pressure Source: PhET Interactive Simulations (English)  Search ARES for similar readings	from Lesson 4, where the lowest hole squirted farthest. They see three identical cans, each with a hole at 10 cm depth, filled to the same level with water, salt water and cooking oil. Individually they write a prediction: which liquid will squirt farthest, or will all be the same? They also predict whether a wide basin and a thin tall tube filled to the same depth would give the same pressure at the bottom, and draw their prediction with a reason. They share in pairs, then with the class, linking to the phenomenon: 'What could make one situation give a larger pressure than another, when something looks the same?'	holes squirted farther. Today all three holes are at the same depth, 10 cm. What do you think will happen? Write your prediction and your reason in your notebook.' WAIT TIME: give 10-15 seconds of silent thinking before anyone speaks. Then cold-call 4 learners from different groups: 'Which liquid did you choose, and why?' Record predictions as a tally on the board (water / salt water / oil / all the same). Do not say who is correct. Ask: 'Mama Wanjiru used the same force with two knives. What do you think is the same and what is different in our three cans?' Accept all answers and record the reasons learners give.	Writing a prediction and a reason first exposes learners' ideas, such as 'a bigger container means more pressure' or 'heavier liquid pushes harder', so the evidence in the Observe Phase can confirm or challenge them. The tally makes the disagreement visible and gives a reason to investigate.	(not just a choice). Look for ideas about liquid type (density) and about container size. Note how many learners predict 'all the same' and how many link depth to pressure from Lesson 4. Listen for misconceptions such as 'more water means more pressure' and record them for later.
Observe Phase  VIDEO: Pressure at a depth in a fluid - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Under Pressure Source: PhET Interactive Simulations (English)  Search ARES for similar readings	In groups of four, learners carry out three short investigations over a tray. Investigation A (density): three identical plastic bottles with a hole at 10 cm (0.10 m) are filled to the same level with water, salt water and cooking oil. Learners observe and measure how far each stream lands on the tray, using a ruler, and record this. They are then given the table of pressures at 10 cm: water 1,000 Pa (1,000 kg/m³), salt water 1,030 Pa (1,030 kg/m³) and cooking oil 920 Pa (920 kg/m³). Investigation B (shape and amount): they look at a demonstration of a wide basin, a narrow tube and a funnel-shaped container, each with water at the same depth of 10 cm and a pressure sensor or a U-tube/pressure gauge at the bottom, and see that the readings are about the same, 1,000 Pa, even though the amounts of water differ a great deal. Investigation C (direction): a bottle with three holes at the same 10 cm depth,	Set up with trays and say: 'Keep the depth the same. Only the liquid changes. Why do we do that?' Cold-call 2 groups to explain fair testing. Circulate and ask: 'Which stream goes farthest? How do you know?' and 'Does 1,030 Pa compared with 920 Pa fit with what you see?' For Investigation B, hold the three containers side by side and ask: 'The wide basin has far more water than the narrow tube. Predict the reading before I show it.' WAIT TIME: 10-15 seconds, then reveal the readings. For Investigation C, ask: 'Does the stream going sideways differ from the one going down? What does that tell us about direction?' Remind groups to wipe spills at once. Ensure each group records data in its table before moving on. Do not give the explanation yet.	Structured data collection with controlled variables. Changing one variable at a time (liquid, container shape, direction) while holding depth constant lets learners isolate which factors matter. Putting the three results side by side in a table helps learners see patterns before any formula, so the conclusions come from their own evidence.	Observe whether groups keep depth at 10 cm and change only one thing. Check that data tables contain correct values with units (Pa) and the order 1,030 Pa > 1,000 Pa > 920 Pa. Look for sketches showing equal streams or readings in all directions. Ask 1 learner per group: 'What did you keep the same, and what did you change?' Note groups that still say the wide basin has more pressure.

	one facing sideways, one angled and one facing down (or a pressure funnel turned in different directions in a bucket), shows streams or readings of the same size in every direction. Learners sketch what they see and record results in a table.			
Explain Phase  VIDEO: Pressure at a depth in a fluid Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Factors Affecting Gas Pressure (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners use their evidence to build explanations in groups. They answer: (1) Which liquid gave the largest pressure at 10 cm and why? They link it to density: salt water (1,030 kg/m ³) is heavier per volume than water (1,000 kg/m ³), and cooking oil (920 kg/m ³) is lighter, so at the same depth pressures are 1,030 Pa, 1,000 Pa and 920 Pa. (2) Why were the readings in the wide basin and narrow tube the same? Because liquid pressure depends on the depth, not on container shape or on the amount of liquid. (3) What does the same stream in every direction tell us? At the same depth the pressure acts equally in all directions. Learners then write a claim-evidence-reasoning statement and compare their ideas with their predictions. They also apply the idea to a Kenyan situation: why a diver in Lake Victoria at Kisumu feels more pressure deeper down, and why a pool of mandazi oil in a sufuria gives less pressure than water at the same depth.	Say: 'Use your table. Write a claim, point to your evidence, and explain why. Use the words depth, density and direction.' WAIT TIME: 10-15 seconds before the first group shares. Cold-call 3 groups to give a claim with evidence, and ask a different learner in each group to add to it: 'Can you build on what your classmate said?' Challenge: 'Someone says the wide basin has more water, so more pressure. Use the evidence to respond.' Ask: 'How is this different from the knives? There the force was the same and the area changed. Here what stays the same and what changes?' Confirm the key ideas: liquid pressure increases with depth and density; it does not depend on container shape or amount of liquid; it acts equally in all directions at the same depth. Do not introduce $P = \rho g h$; tell learners that Lesson 6 will show how to calculate it.	Claim-Evidence-Reasoning (CER) with peer challenge. Learners must tie each claim to specific numbers (1,000 / 1,030 / 920 Pa) and observations. Arguing against the idea that 'more water means more pressure' replaces a misconception with evidence-based reasoning. Comparing with the knife phenomenon helps learners see that pressure depends on specific quantities.	Listen for correct use of 'depth' and 'density' and for statements that container shape and amount of liquid do not change pressure at a given depth. Check CER statements for a claim, a number with its unit, and a reason. Indicators of mastery: states that salt water gives 1,030 Pa and oil 920 Pa at 10 cm because of density; says pressure is equal in all directions at the same depth. Note any learner who still says direction changes the pressure and revisit Investigation C with them.
Driving Question Board (DQB) Creation  VIDEO: Pressure at a depth in a fluid - dubbed(KY)	Learners return to the Driving Question Board with the central question about why a sharp knife cuts better than a blunt knife with the same force. They place sticky notes on the board in three ways: (a) mark questions that Lessons 4 and 5 have partly answered, such	Say: 'Look at our board. Which questions can we now answer with evidence from today?' WAIT TIME: 10-15 seconds, then cold-call 3 learners to move a note and justify it. Ask: 'What new question has today given you?' Invite 3 more learners to add notes. Point to	Driving Question Board as a public record of growing understanding. Moving questions between 'answered', 'partly answered' and 'new' shows learners how the evidence is building towards the central question and gives them ownership of the learning path.	Check that notes are specific and related to pressure (not vague). Indicators: at least two new questions per group about calculation or other fluids; correct sorting of answered questions with a reason. Note learners who link liquid pressure back to the knife

Source: Khan Academy (Kiswahili) Search ARES for similar videos EXERCISE: Factors Affecting Gas Pressure Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	as 'Does the depth matter in liquids?' and 'Does the liquid type matter?'; (b) add new questions that arose, for example 'How can we calculate the pressure in a liquid?', 'Does air also press on us like water does?', and 'Why do dams need thicker walls at the bottom?'; (c) connect questions to the knives, for example 'Is pressure in the knife also a matter of just one quantity?' They sort the questions under the KICD inquiry questions, such as 'What factors affect pressure in fluids?' and 'Why does pressure increase with depth?'. Learners return to their group model of how force, area and pressure explain the knives and add a new section for liquids. On a large sheet they draw: a container of liquid with arrows of equal length pushing outwards in all directions at the same depth; longer arrows lower down to show greater depth; and a note to show that denser liquid (salt water) has longer arrows than less dense liquid (oil) at the same depth. Beside it they list 'what matters': depth and density; 'what does not matter': container shape and amount of liquid. They also label how this fits with the knife model: pressure = force / area in solids (same force, different area), and in liquids the pressure depends on depth and density. Groups mark parts they are unsure of, such as 'how to calculate it', to be answered in Lesson 6.	links: 'Today we learned liquid pressure depends on depth and density. Which question about numbers will we answer next lesson?' Highlight the question about calculating liquid pressure as the focus for Lesson 6. Keep the note that the sharp knife question is still not fully answered because the class has not yet pulled all the ideas together.		question.
Model Building Phase VIDEO: Pressure and Pascal's principle (part 1) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Pressure and Density of the Atmosphere (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings		Say: 'Add liquids to your model. Use arrows to show the pressure at different depths and in different directions. Label what matters and what does not.' Circulate and ask: 'Where is the pressure greater, near the surface or deep down? How does your arrow show that?' and 'What would change in your model if the liquid were oil?' WAIT TIME: 10-15 seconds after each question. Cold-call 2 groups to present. Ask: 'How does this connect to the sharp and blunt knife? What is similar and what is different?' Encourage groups to mark in a different colour what they changed from earlier versions and why. Collect the models to review before Lesson 6.	Cumulative model revision. Learners extend the same model across lessons, adding new evidence and marking unresolved questions. Drawing arrows for depth, density and direction turns the ideas into a visual form that can later be linked to the formula $P = \rho g h$.	Check that models show equal arrows in all directions at the same depth, longer arrows at greater depth, and a note that density changes the pressure. Look for correct labelling of container shape and amount of liquid as not mattering. Indicators of strong models: a link to the knives (same force, different area) and a statement of what remains to be explained. Note learners whose models show pressure only downwards.

D. TEACHER REFLECTION

After the lesson ask yourself: Did learners keep depth constant when comparing liquids, or did some groups change two things at once? How many still believed that more water or a wider container means more pressure after Investigation B, and what evidence moved them? Did learners notice that pressure is equal in all directions at the same depth, or did they focus only on downward pressure? Were the salt water and oil differences (1,030 Pa and 920 Pa compared with 1,000 Pa) too small for learners to see in the streams, and would readings from a gauge or a table help more? Which learners connected today's work to the knives and which need more support? Note the questions on the DQB to carry into Lesson 6, and prepare extra examples with Lake Victoria and dams for learners who need them.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that at the same depth of 10 cm, salt water (1,030 Pa) gave a larger pressure than water (1,000 Pa), and cooking oil (920 Pa) gave a smaller pressure. We saw that containers of different shapes and amounts of water gave the same pressure at the same depth, and that streams from holes at the same depth were equal in all directions.
What did I learn?	We learned that liquid pressure increases with depth and with density, but does not depend on container shape or the amount of liquid. At the same depth the pressure acts equally in all directions.
How does this explain the phenomenon?	This helps explain the knife question by showing that pressure depends on particular quantities, not on how large or heavy something looks. For the knives, the force (40 N) is the same and the area is different, so the pressure is different. In liquids, depth and density decide the pressure. We still need to learn how to calculate liquid pressure and bring all the ideas together to fully explain why the sharp knife cuts.

LESSON 6 (40 minutes (one lesson; or 2 x 40 minutes if the derivation is done at a relaxed pace)): Applying $P = \rho gh$: Deriving the Liquid Pressure Equation and Calculating Pressure in Lake Victoria and Dams

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To help learners derive $P = \rho gh$ from $P = F/A$ by thinking of the weight of a column of liquid pressing on its base, and to use the equation to calculate liquid pressure at given depths in Lake Victoria and a dam. This extends the pressure = force / area idea that explains why the sharp knife cuts (same force, different area) to liquids.
Knowledge	By the end of the lesson, the learner should be able to: (a) explain that the pressure at a depth in a liquid comes from the weight of the liquid column above it; (b) derive $P = \rho gh$ using $P = F/A$, $W = mg$ and $m = \rho V$; (c) state that liquid pressure depends on density ρ and depth h , not on the area of the column; (d) calculate liquid pressure using $P = \rho gh$, with $g = 10 \text{ N/kg}$, giving answers in Pa and kPa.
Skills	Developing and using models (a liquid column model of Lake Victoria); Using mathematics and computational thinking (deriving and applying $P = \rho gh$, converting Pa to kPa); Analysing and interpreting data (pressure vs depth for Lake Victoria); Constructing explanations (why the dam wall is thicker at the bottom); Communication and collaboration (think-pair-share, group derivation).
Attitudes	Responsibility: learners take different roles (recorder, calculator, reporter) when working in groups. Respect: learners listen to and value different ideas during the derivation. Appreciation of the use of physics in the design of Kenyan dams such as Masinga and in understanding Lake Victoria.
Key Inquiry Question	KICD Key Inquiry Questions addressed: 'What factors affect pressure in fluids?' and 'Why does pressure increase with depth?' Lesson question: How can we calculate the pressure at any depth in Lake Victoria or behind a dam, and how does this connect to pressure = force / area, which explains why a sharp knife cuts better than a blunt knife when using the same force?
Purpose in Storyline	Lesson 2 gave us $P = F/A$. Lesson 5 showed that liquid pressure depends on depth and density. Lesson 6 joins these ideas: the force on an area under water is the weight of the liquid above it, giving $P = \rho gh$. The learners get a calculating tool that is used in Lesson 7 (the atmosphere as a column of air) and Lesson 9 (dam 30 m deep). For the knife phenomenon, it shows that $P = F/A$ is a general idea, not only for knives and bricks: the force is always spread over an area.
Safety Notes	Keep water spills wiped up to prevent slipping. If a transparent tube or bottle column demonstration is used, place it on a tray. Learners should not climb on desks or stools to read the height of the column. Remind learners never to swim in deep lakes or dams without supervision; real water pressure at depth is dangerous.

B. LESSON OVERVIEW

Learners begin by predicting how the pressure at the bottom of Lake Victoria changes as they go deeper and whether the width of the lake matters. In the Observe Phase they use a model of a column of water (a tall transparent container or paper drawing with a 1 m^2 base) and the data from earlier lessons (brick on flour, can with holes) to build up the derivation step by step: $P = F/A$, the force is the weight $W = mg$, mass $m = \rho V$, and $V = Ah$, so $P = \rho gh$. They then calculate pressures at 10 m, 20 m and 40 m in Lake Victoria (100 kPa, 200 kPa, 400 kPa) and at the base of a 30 m dam (300 kPa). In the Explain Phase they link the result back to the knife: $P = F/A$ always needs force and area, and just as the knife edge changes area while the force stays the same (40 N), in liquids the area cancels, leaving depth and density as the only factors. The lesson ends with updating the DQB and the class model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Pressure at a depth in a fluid Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Pressure and Density of the Atmosphere (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners look at a sketch of Lake Victoria near Kisumu with three marked points at depths of 10 m, 20 m and 40 m. Individually, for 2 minutes, they write a prediction in their notebooks: (1) What happens to the pressure as we go from 10 m to 20 m to 40 m? (2) If the depth doubles, what happens to the pressure? (3) If Lake Victoria were twice as wide, would the pressure at 20 m change? Why? They then share with a partner and compare. Finally, they recall the knife: same force of 40 N, different area, different pressure, and consider whether liquids work the same way.	Display the sketch on the board. Say: 'Imagine you are a diver in Lake Victoria off Kisumu. Your ears hurt as you go deeper. What is changing?' Give WAIT TIME of 10-15 seconds after each question before taking answers. Say: 'Write your prediction first, then talk to your partner. I want a number or a pattern, not just "it gets bigger".' Cold-call 4 learners with different predictions (include at least one who says the lake width matters) and record them on the board in a 'Predictions' column without saying right or wrong. Say: 'Today we will see whether we can calculate the answer instead of guessing.'	Predict-Observe-Explain (POE) with Think-Pair-Share: committing to a written prediction makes learners aware of their prior ideas (for example 'wider lake means more pressure'), so the later evidence can confront them. Linking to the knife keeps the pressure = force / area idea in view.	Check the notebooks for: (a) a written prediction that pressure increases with depth (from Lessons 4-5); (b) whether learners suggest a numerical pattern such as doubling; (c) whether learners claim width or amount of water matters. Tally on the board how many hold the 'width matters' idea (about 3 or 4 learners is typical). Observable indicator: learners who write 'pressure doubles when depth doubles' show readiness to move to the derivation.
Observe Phase  VIDEO: Pressure at a depth in a fluid - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Pressure and Density of the Atmosphere Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Part A: Building the water column (15 minutes). In groups of 4, learners get a drawing (or a transparent container) showing a column of water of base area A above a point at the bottom, with height h. They work through the derivation in steps on a worksheet, using a 1 m ² column of water first so the numbers are easy: Step 1: A column of water 10 m high with a base of 1 m ² has volume $V = 1 \times 10 = 10 \text{ m}^3$. Step 2: Mass $m = \rho V = 1,000 \times 10 = 10,000 \text{ kg}$. Step 3: Weight $W = mg = 10,000 \times 10 = 100,000 \text{ N}$. Step 4: Pressure $P = F/A = 100,000 / 1 = 100,000 \text{ Pa} = 100 \text{ kPa}$. They repeat the steps for a base of 2 m ² ($F = 200,000 \text{ N}$, $A = 2$, $P = 100,000 \text{ Pa}$) and notice that the pressure is unchanged. Then they do it with letters: $V = Ah$, $m = \rho Ah$, $W = \rho Ahg$, $P = W/A =$	Circulate with the worksheet answers. Say: 'We know $P = F/A$ from the brick and the knife. What is the force on the bottom of this water column? Think about what is pressing down.' WAIT TIME 10-15 seconds, then cold-call 3 groups for the force (the weight of the water). Say: 'Weight is $W = mg$. But we do not know the mass. What do we know about the water that lets us find mass?' WAIT TIME 10-15 seconds. Guide to density: $m = \rho V$. When groups compare 1 m ² and 2 m ² bases, ask: 'The 2 m ² column holds twice as much water and twice the weight. Why is the pressure the same?' Cold-call 2 groups. Ask: 'Which letter cancelled out?' (A). Stress units: 'Always write the unit: $\text{kg/m}^3 \times \text{N/kg} \times \text{m}$ gives N/m^2 , which is Pa.' For Part B, say:	Scaffolded derivation with a concrete-to-abstract progression (numbers first, then letters): learners first work with a 1 m ² column, where each step can be checked, then generalise to symbols. Comparing the 1 m ² and 2 m ² columns shows that area cancels, building understanding of why only ρ and h matter. Data table and graph: patterns in the Lake Victoria data (100, 200, 400 kPa) show direct proportionality between P and h.	Observable indicators: (a) groups correctly identify the weight of the water column as the force; (b) groups write the chain $V = Ah$, $m = \rho Ah$, $W = \rho Ahg$, $P = \rho gh$ without help by the end of the activity; (c) the table is filled in with correct values (100 kPa, 200 kPa, 400 kPa, and 300 kPa for the dam) with units; (d) learners state that the area cancels. Look for common errors: forgetting to convert to Pa/kPa, using $g = 9.8$ instead of 10, mixing up cm and m. Ask 3 groups to show their worked solutions on the board.

	pgh. Part B: Using the equation (10 minutes). Learners calculate the pressure from water alone in Lake Victoria ($\rho = 1,000 \text{ kg/m}^3$, $g = 10 \text{ N/kg}$): at $h = 10 \text{ m}$: $P = 1,000 \times 10 \times 10 = 100,000 \text{ Pa} = 100 \text{ kPa}$; at $h = 20 \text{ m}$: $200,000 \text{ Pa} = 200 \text{ kPa}$; at $h = 40 \text{ m}$: $400,000 \text{ Pa} = 400 \text{ kPa}$. They fill in a table and plot a quick graph of P against h to see a straight line through the origin. Then they calculate the pressure at the base of a dam with water 30 m deep: $P = 1,000 \times 10 \times 30 = 300,000 \text{ Pa} = 300 \text{ kPa}$.	'Check your arithmetic with your partner. Convert to kPa by dividing by 1,000.' Use the stopping question: 'If depth doubles from 10 m to 20 m, what happens to the pressure? Does your table agree with your prediction?' Address the lake-width idea directly: 'Your prediction said a wider lake gives more pressure. What does the equation say?'		
Explain Phase  VIDEO: Pressure at a depth in a fluid Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Pressure and Density of the Atmosphere (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners work as a whole class and then in pairs to explain and apply the results. Task 1: Using $P = pgh$, explain in 3 sentences why the wall of a dam like Masinga is built thicker at the bottom than at the top (pressure at the bottom of a 30 m dam is 300 kPa, which is three times the 100 kPa at 10 m). Task 2: Explain why the pressure does not depend on the width of Lake Victoria, using the area cancelling in the derivation. Task 3 (connecting to the knife): Complete the sentence frame: 'For the knives, the force was the same (40 N) but the area was different, so the pressure was different. For the water column, the area cancels because _____, so the pressure depends only on _____ and _____.' Task 4 (short practice): Calculate the pressure from the water alone at 15 m depth in Lake Victoria (150,000 Pa = 150 kPa). Then calculate for cooking oil ($\rho = 920 \text{ kg/m}^3$) at 10 m depth (92,000 Pa = 92 kPa) to practise changing density.	Say: 'Turn to your partner and explain to each other why the dam wall is thicker at the bottom. Use the number 300 kPa in your explanation.' WAIT TIME 10-15 seconds before pairs begin. Cold-call 3 pairs. Push for reasoning: 'You said the pressure is bigger at the bottom. Where did that number come from? Show me using the equation.' For the knife link, say: 'The knives used $P = F/A$ with different areas. Here the area cancels. Is the idea of pressure still force per area? Yes, because the force is the weight of water spread over the area.' Do not re-teach the knife calculations; refer only to the idea of same force and different area. Cold-call 2 learners to complete the sentence frame aloud. For the practice task, walk around and check units. Address misconceptions: 'Does a bucket of water at 10 m depth in a small pond give the same pressure as in Lake Victoria at 10 m? What does the equation say?'	Claim-Evidence-Reasoning (CER) with sentence frames: learners make a claim (the dam wall must be thicker at the bottom), support it with evidence (300 kPa vs 100 kPa from calculations) and reason from the equation ($P = pgh$, so P rises with h). The sentence frame linking back to the knife builds a bridge between the new idea and the anchoring phenomenon, so learners see $P = F/A$ as one idea in two situations.	Observable indicators: (a) explanations mention the pressure at the base increases with depth and give numbers with units; (b) learners state that width or total amount of water does not change the pressure at a given depth; (c) the sentence frame is completed with 'the area cancels' and 'depth and density'; (d) the practice answers 150 kPa and 92 kPa are correct. Use a thumbs-up/sideways/down self-check on confidence with the equation, and plan a quick re-teach of the step '$m = \rho V$' for learners showing thumbs-down.
Driving Question	Learners return to the Driving Question Board started in Lesson	Say: 'Look at our DQB. Which questions can we now answer with	Driving Question Board as a living record: moving questions to	Observable indicators: (a) learners move the correct questions (for

Board (DQB) Creation  VIDEO: Pressure at a depth in a fluid - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Pressure and Density of the Atmosphere (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	1. In groups, they look at the questions on the board and move at least one sticky note to an 'Answered', 'Partly answered' or 'Still to investigate' area. They write one new question on a sticky note that arose today, for example: 'Does the air above the lake add to the pressure?', 'Does the pressure act sideways at depth?' or 'Is the knife like the water column at all?' They add one entry to the class 'What we learned' column such as: 'Liquid pressure $P = \rho gh$ depends on density and depth.'	a number, using what we found today?' WAIT TIME 10-15 seconds. Cold-call 3 learners to move notes and justify. Say: 'Our main question is why a sharp knife cuts better than a blunt knife with the same force. How does today connect?' Cold-call 2 learners (expected answer: pressure is force per area, and the sharp knife has a smaller area, so greater pressure). Prompt: 'Today the area cancelled for water. For the knife the area did not cancel. Why not?' (The knife pressure depends on the area of contact; the water pressure is from weight spread over the same area as the column.) Collect new questions: 'Hold up your new question; I will read three of them.' Highlight a question about the air above the lake and say it will be investigated in Lesson 7.	'answered' categories shows learners their own progress, and new questions create a purpose for the next lesson. Tying today's learning back to the knife keeps the anchoring phenomenon central.	example 'Why does pressure increase with depth?') to 'Partly answered' or 'Answered' with a reason; (b) at least 5 new questions are produced, and some link to the atmosphere or the knife; (c) learners can state in one sentence how today's learning relates to the knife question. Note which groups still ask whether the width of the lake matters, as it may signal a lingering misconception.
Model Building Phase  VIDEO: Pressure and Pascal's principle (part 1) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Pressure and Density of the Atmosphere Practice <i>Related topic: not an exact match for this lesson.</i>	Learners revise their class and individual models of how pressure works. In their notebooks they draw two linked diagrams: (1) the knife on a tomato: label force 40 N, small contact area, large pressure (idea only; numbers come from earlier lessons), and (2) a column of water above a point in Lake Victoria: label depth h, density ρ, weight of the column $W = mg$ pressing down on area A, and the equation $P = \rho gh$. They add an arrow showing 'pressure = force / area in both' and a short caption: 'Same idea: pressure is force on an area. Knife: change the area. Liquid: the weight of the column depends on depth and density.' They add a small data box: Lake Victoria 10 m = 100 kPa, 20 m = 200 kPa, 40 m = 400 kPa, dam 30 m = 300 kPa.	Say: 'Open your model from the last lesson. What can we add today that was missing?' WAIT TIME 10-15 seconds. Cold-call 3 learners to suggest additions. Say: 'Draw the water column and label every part of the equation. If I point to ρ, what does it tell me about the liquid? If I point to h, what does it measure?' Cold-call 2 learners. Circulate and check that learners show h measured vertically from the surface (not along a slope) and the units in Pa/kPa. Ask 2 learners to show their models under the document camera or on the board for a gallery check. Close with an exit ticket: 'Calculate the pressure from water alone at 25 m depth in Lake Victoria (250,000 Pa = 250 kPa) and write one sentence linking $P = F/A$ to $P = \rho gh$.' Collect the exit tickets.	Cumulative model revision: learners add a new component (the liquid column) to their existing model, so the model gets more complete and links two situations (solid contact and liquids) through the one idea of pressure = force / area. Writing a short caption forces learners to explain relationships, not just list facts.	Observable indicators in models: (a) a labelled water column with h, ρ, W and A; (b) the equation $P = \rho gh$ with correct units; (c) h measured vertically from the surface; (d) a link statement connecting liquids to $P = F/A$ and to the knife idea of same force, different area. Exit ticket: the correct answer is 250 kPa; score 1 for the number, 1 for units, 1 for the link sentence. Learners scoring 2 or less get a short re-teach of $m = \rho V$ and unit conversion at the start of Lesson 7.

Source: CK-12 Search ARES for similar readings				
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D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners understand why the force on the base of the column is the weight of the liquid (not something extra)? Which groups needed prompts on $m = \rho V$? (2) How many learners still believe the width of Lake Victoria or the total amount of water changes the pressure at a given depth, and what evidence (for example the 1 m² vs 2 m² comparison) changed their minds? (3) Were learners careful with units and with $g = 10 \text{ N/kg}$ (not 9.8)? (4) Did the connection to the knife feel natural, or did it seem like a separate topic? How could the same force, different area idea be reinforced in Lesson 7? (5) Was the WAIT TIME of 10-15 seconds used consistently, and did more learners participate in cold-calls than before? Note questions on the DQB that will be used to shape Lesson 7 (atmospheric pressure).

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We worked out the pressure of water at different depths: Lake Victoria 10 m = 100 kPa, 20 m = 200 kPa, 40 m = 400 kPa, and the base of a dam 30 m deep = 300 kPa. When the column base was 1 m ² or 2 m ² , the pressure was the same at the same depth.
What did I learn?	Liquid pressure comes from the weight of the liquid above. Using $P = F/A$, $W = mg$, $m = \rho V$ and $V = Ah$, we get $P = \rho gh$. Pressure depends on the density ρ and depth h , not on the width of the lake or the amount of water. Pressure in kPa = pressure in Pa divided by 1,000.
How does this explain the phenomenon?	The pressure in a liquid rises with depth because a deeper column has more weight pressing on each square metre. This is why a dam wall is thicker at the bottom (300 kPa at 30 m compared with 100 kPa at 10 m). It is the same idea as the knives: pressure is force on an area; with the knives the force (40 N) was the same but the area was different, so the pressures were different.

LESSON 7 (40 minutes): Atmospheric Pressure and Its Effects

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use evidence from four demonstrations and Kenyan altitude data to explain that air has weight and so exerts atmospheric pressure (about 101 kPa at sea level) that decreases with altitude, and they link this to $P = F/A$, which also explains Mama Wanjiru's knives.
Knowledge	Air is a fluid with weight. The atmosphere is a column of air pressing on every surface, so P_{atm} is about 101 kPa (101,000 Pa) at sea level (Mombasa). P_{atm} decreases with altitude: Mombasa 101 kPa, Kisumu 89 kPa, Nairobi 83 kPa, Mount Kenya summit 54 kPa. Force from air = $P_{\text{atm}} \times A$. For the glass mouth of 0.004 m ² , the upward force is 101,000 x 0.004 = 404 N, much more than the 2.5 N weight of 250 g of water, so the cardboard stays. Effects: inverted glass, crushed can, ears popping, drinking straw.
Skills	Predicting and observing; analysing and interpreting data (Mombasa, Kisumu, Nairobi, Mount Kenya); using mathematics ($F = P \times A$); developing and using models (drawing force arrows); constructing explanations from evidence; engaging in argument from evidence.
Attitudes	Curiosity about invisible air; Respect for peers' ideas; Responsibility in handling glass and hot equipment; Appreciation of how physics explains everyday Kenyan experiences. Core competencies: Critical thinking and problem solving; Communication and collaboration.
Key Inquiry Question	What is atmospheric pressure? (KICD Key Inquiry Question 5). Supports the driving question: Why does a sharp knife cut better than a blunt knife when using the same force?
Purpose in Storyline	Lessons 2 and 3 taught $P = F/A$ and the effect of area. Lesson 6 showed pressure from the weight of a column of liquid. Lesson 7 extends the column idea to air, so students see that pressure can come from a column of fluid and that $P = F/A$ works on huge areas such as a glass mouth. Students also notice that the same air pressure acts on both knives, so atmospheric pressure does not explain the difference between them. The difference is still the area (same force, different area). This prepares the siphon in Lesson 9 and the hydraulics in Lesson 8.
Safety Notes	Use plastic cups or thick glasses over a tray or basin. Never use cracked glass. For the crushed can, the teacher alone heats a small amount of water in an empty soda can on a jiko or spirit burner and uses tongs; keep students 2 m back and wear no loose clothing near the flame. Hot steam and hot cans can burn. Mop spills immediately. Do not suck hard on straws shared between learners; each learner uses a clean straw.

B. LESSON OVERVIEW

Mama Wanjiru's kitchen in Kisumu is full of air she cannot see. In this lesson learners predict what happens when a glass full of water is turned upside down over cardboard, then observe four effects of air: the inverted glass, the crushed can, ears popping on a trip up from Mombasa toward Nairobi, and drinking through a straw. They analyse Kenyan altitude data (Mombasa 101 kPa, Kisumu 89 kPa, Nairobi 83 kPa, Mount Kenya 54 kPa), calculate the air force on the glass mouth (404 N against the 2.5 N weight of water), and build the idea that air has weight and presses on all surfaces. They then add this to the class model and DQB, noting that the air pushes equally on both knives, so the knife difference remains the area.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Pressure and Pascal's principle (part 1) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Air Pressure and Altitude (Flexbook) Source: CK-12  Search ARES for similar readings	Learners see a plastic cup filled to the brim with water (250 g) and a piece of cardboard on top. Individually (2 min) they draw what they think will happen when the cup is turned upside down while the cardboard is held, then the hand is removed. They write one sentence starting 'I think ... because ...'. They then share with a partner. Link to the phenomenon: Mama Wanjiru also pushes down with a force; here, what pushes up on the cardboard?	Hold up the cup and say: 'Mama Wanjiru fills this cup at her kitchen tap in Kisumu. If I turn it upside down, what will happen to the water and the cardboard? Draw it and give a reason.' WAIT TIME: 15 seconds of silent thinking before any talk. Then say: 'Turn to your partner and compare.' Cold-call 4 learners (2 who predict it falls, 2 who predict it stays) and ask each: 'What force do you think acts on the cardboard, and in which direction?' Record predictions in two columns on the board without judging. Do not reveal the result yet.	Predict-Observe-Explain (Prediction with reasoning). Committing to a prediction and a reason exposes the common idea that air is 'nothing' and creates a need to know. It lets the teacher compare later evidence against learners' own ideas.	Look for: (1) drawings that show the water and cardboard falling, or staying; (2) whether learners draw any upward arrow, and if so whether it is labelled as air; (3) whether they say 'air has no weight' or 'air does not push'. Tally on a clipboard: ____ say it falls, ____ say it stays, ____ mention air. At least 4 cold-called learners should give a reason, not just an answer.
Observe Phase  VIDEO: Pressure and Pascal's principle (part 1) - dubbed(KY) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Air Pressure and Altitude Practice Source: CK-12  Search ARES for similar readings	Stations or teacher demonstrations (about 15 minutes in total), with groups of 4 recording in a table (what we did / what we saw / possible reason). 1) Inverted glass: the cup is turned over the cardboard; the water stays. Learners tilt the cup slowly sideways and see the card fall when air gets in. 2) Crushed can (teacher demo): a little water boiled in an empty soda can, can quickly inverted into a basin of cold water; it collapses with a bang. 3) Ears popping: learners recall travelling by bus or matatu from Mombasa to Nairobi or up to Kericho, and ears popping; they read the data card. 4) Drinking straw: learners drink water or uji through a straw, then place a finger on top of the straw and lift it out, noting liquid stays in. Data card: Mombasa 101 kPa, Kisumu 89 kPa, Nairobi 83 kPa, Mount	Say: 'Watch the cardboard closely. Write what you see, not what you think.' At the glass, WAIT TIME 10 seconds after the cup is turned over before asking anything. Ask: 'What is holding the card up? Where could the force come from?' For the can say: 'Predict first: will the can stay the same, swell or collapse?' Cold-call 3 learners for predictions, then demonstrate with tongs from a safe distance. Circulate during the straw activity and ask: 'When you put your finger on top, what stops the liquid from falling out?' For the data card ask: 'What pattern do you see as we go from Mombasa to Mount Kenya? Tell your partner in one sentence.' Cold-call 3 pairs. Press for evidence: 'Which number shows that?' Keep a record of learners' phrases such as 'air pushes' for later.	Gathering evidence across multiple phenomena and pattern spotting (Analysing and interpreting data). Four different effects and one data table push learners to look for a common cause rather than explaining each separately. Graphing altitude against pressure makes the trend visible.	Observable indicators: (1) group tables contain at least three correct observations (card stays; can collapses; pressure falls with height: 101, 89, 83, 54 kPa); (2) learners use words like 'push', 'air' and 'outside' rather than 'suction pulls'; (3) learners can state the trend in one sentence: 'Higher up, the air pressure is lower.' Note groups still writing 'the can sucked the water' and prompt them with: 'What was pushing from outside?'

	Kenya summit (5,199 m) about 54 kPa. Learners plot altitude against pressure on a quick sketch graph.			
Explain Phase  VIDEO: Pressure and Pascal's principle (part 1) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Atmospheric Pressure (Flexbook) Source: CK-12  Search ARES for similar readings	Learners make sense of the evidence in three steps. Step 1 (5 min): Think of the atmosphere as a very tall column of air above us, like the water column in Lesson 6, pressing down because air has weight. They draw the glass with arrows: air pushing up on the cardboard from outside and water weight pushing down. Step 2 (5 min): Calculate the force of air on the glass mouth: mouth area $40 \text{ cm}^2 = 0.004 \text{ m}^2$; $F = P \times A = 101,000 \text{ Pa} \times 0.004 \text{ m}^2 = 404 \text{ N}$ upward. Compare with the water weight: $250 \text{ g} = 0.25 \text{ kg}$, $W = 0.25 \times 10 = 2.5 \text{ N}$. 404 N is far greater than 2.5 N, so the cardboard stays. Step 3 (5 min): Use the idea to explain the crushed can (steam pushed air out; when cooled, inside pressure falls and the outside air pushes the can in), ears popping (air pressure drops as altitude rises, so the higher pressure trapped in the middle ear pushes out until it equalises), and the straw (sucking lowers the pressure in the straw and the atmospheric pressure on the liquid surface pushes the liquid up). Learners then answer: Does the air press equally on Mama Wanjiru's sharp and blunt knives? What actually differs between them?	Say: 'We said in Lesson 6 that a column of water presses because of its weight. Why might a column of air do the same?' WAIT TIME 10 seconds. Cold-call 3 learners. Write on the board: $P = F/A$, so $F = P \times A$. Guide the calculation on the board with learners supplying each line: 'Mouth area in cm^2? Convert to m^2 ($1 \text{ cm}^2 = 0.0001 \text{ m}^2$). Then $F = 101,000 \times 0.004$.' Ask: 'Compare 404 N with 2.5 N. What does that tell us about the cardboard?' Cold-call 2 learners. Address the misconception: 'Suction does not pull. What is pushing?' For the straw, ask: 'In Kisumu, would drinking through a straw work the same as in Mombasa? Why or why not?' Then bridge: 'Air presses on both knives with the same pressure. So does air explain why one knife cuts? Talk to a partner.' WAIT TIME 15 seconds, then cold-call 3 pairs. Confirm: same force (40 N), different area; atmospheric pressure is not the difference.	Claim-Evidence-Reasoning with mathematical reasoning and a force-arrow diagram. Learners must state a claim (air presses), support it with evidence (404 N vs 2.5 N, collapsing can, falling pressures with height), and reason with $P = F/A$. Reusing $P = F/A$ on a new surface connects this lesson to the knives.	Check: (1) learners convert 40 cm^2 to 0.004 m^2 correctly; (2) $F = 404 \text{ N}$ is calculated with the unit; (3) learners explain that the air force is greater than the water weight, so the card stays; (4) learners explain the can or straw using 'outside air pushes' language; (5) learners say that air pressure acts on both knives equally so area is still the key difference. Exit question on a slip: 'Why does ear popping happen more going from Mombasa to Nairobi?' Look for 'pressure outside is lower at altitude'.
Driving Question Board (DQB) Creation  VIDEO: Pressure and Pascal's principle	Groups revisit the Driving Question Board. They (1) place a sticky note answering a question on the DQB with new evidence (for example, 'What is pressure? Atmospheric pressure is pressure from the weight of air: about 101 kPa at	Say: 'Take two minutes. What new question has today's lesson raised about pressure? Write it on a sticky note.' WAIT TIME 10 seconds before they start writing. Cold-call 4 groups to read one note aloud and say where it goes	Driving Question Board (question tracking). Revisiting questions shows what has been answered and what is still open, keeping learners as owners of the inquiry and tying the new topic to the anchoring phenomenon.	Count notes: at least 6 new questions or answers posted. Look for whether questions go beyond 'what' to 'why' and 'how', and whether at least 2 groups connect atmospheric pressure back to the knives. Note groups whose

(part 1) - dubbed(KY) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili) Search ARES for similar videos EXERCISE: Atmospheric Pressure Practice Source: CK-12 Search ARES for similar readings	Mombasa'); (2) add new questions that arose, such as 'Why is it harder to drink uji through a straw on Mount Kenya?' or 'Can pressure be moved from one place to another, like in a hydraulic jack?'; (3) mark which KICD key inquiry question (5) has been answered in part. They also write one sentence on how this lesson relates to the knives.	on the board. Ask: 'Does this help answer why the sharp knife cuts better? Or is it a new thread?' Place questions about transmitting pressure in a cluster for Lesson 8 and the siphon and straw in a cluster for Lesson 9. Praise specific connections: 'Your question about Mount Kenya links to our altitude data.'		questions show a lingering misconception (for example 'air does not weigh anything') and plan a follow-up.
Model Building Phase VIDEO: Pressure and Pascal's principle (part 1) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Air Pressure and Altitude (Flexbook) Source: CK-12 Search ARES for similar readings	Learners return to the class model from earlier lessons, which has: same force of 40 N, area, and $P = F/A$ explaining the knives. They add a new section: 'Fluids press on surfaces.' They add a diagram of the atmosphere as a column of air above a surface with arrows, plus the numbers 101 kPa (Mombasa), 89 kPa (Kisumu), 83 kPa (Nairobi) and 54 kPa (Mount Kenya). They revise the model with an annotation: 'Air presses equally on both knives; the difference between knives is still area (same force, different area), not air.' They write a summary statement in their notebooks: 'Air has weight, so it exerts atmospheric pressure of about 101 kPa at sea level, and this decreases with altitude.'	Say: 'Open the class model. What do we need to add after today to show what air does?' WAIT TIME 15 seconds. Cold-call 3 learners to suggest additions and ask each: 'What evidence from today supports that?' Add the atmospheric column diagram with learners' input. Then ask: 'Which parts of the model explain the knives, and which parts explain other situations such as the glass, the can and the straw?' Cold-call 2 pairs. Make sure the model keeps $P = F/A$ as the central relationship. Finally say: 'For Lesson 8, think about how pressure applied in one place could move through a fluid. Bring one idea.'	Cumulative model revision (Developing and using models). Adding atmospheric pressure to the existing model lets learners see how a single relationship, $P = F/A$, explains varied phenomena, and clarifies what the model does not yet explain.	Check that each learner's model includes: (1) a column of air with arrows; (2) P_{atm} values for at least two Kenyan places; (3) the statement that pressure decreases with altitude; (4) a note that the knives differ in area, not in air pressure. Use a quick thumbs check: 'Can you explain the inverted glass using $P = F/A$?' and select 3 learners to explain in 30 seconds.

D. TEACHER REFLECTION

Which learners still describe 'suction' as a pulling force rather than outside air pushing? Did the 404 N versus 2.5 N comparison make the idea of atmospheric pressure convincing, and did learners keep unit conversions (cm² to m²) correct? Did every group connect atmospheric pressure back to the knives and recognise that the difference between them is still area? What evidence from the QJB suggests learners are ready for transmission of pressure in fluids in Lesson 8?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	The water stayed in the inverted glass with the cardboard; the heated can collapsed when cooled in water; the straw held liquid when the top was closed; ears pop when travelling to higher places such as Nairobi; atmospheric pressure values fall from Mombasa 101 kPa to Kisumu 89 kPa, Nairobi 83 kPa and Mount Kenya 54 kPa.
What did I learn?	Air has weight, so it exerts atmospheric pressure, about 101 kPa (101,000 Pa) at sea level, and it acts on every surface. Atmospheric pressure decreases with altitude. The air force on the glass mouth is $F = P \times A = 101,000 \times 0.004 = 404 \text{ N}$, much more than the 2.5 N water weight. Air does not make one knife cut better than the other because it pushes equally on both.
How does this explain the phenomenon?	The cardboard stays because the upward force of air (404 N) is much greater than the weight of the water (2.5 N). The can collapses because the pressure inside falls and outside air pushes it in. Ears pop because outside pressure falls as altitude rises. The straw works because atmospheric pressure pushes liquid up when pressure in the straw is lowered. For the knives, the same force (40 N) acts on different areas, so the difference is area, not air.

LESSON 8 (40 minutes (single lesson)): Transmission of Pressure in Fluids: Pascal's Principle in Syringes, Hydraulic Jacks and Brakes

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To help learners discover that pressure applied to an enclosed fluid is transmitted equally in all directions, so a small force on a small piston produces a larger force on a larger piston ($F_1/A_1 = F_2/A_2$), and to connect this to the knife phenomenon: pressure = force / area, where the area is what changes the effect.
Knowledge	By the end of the lesson, the learner should be able to: (a) state that pressure applied to an enclosed fluid is transmitted equally in all directions (Pascal's principle); (b) use $F_1/A_1 = F_2/A_2$ to calculate the load force from the hydraulic jack data (5 cm ² and 100 cm ² pistons, 50 N effort, 100 kPa, 1,000 N load); (c) explain how syringes, hydraulic jacks and brakes use this principle.
Skills	Predicting from prior knowledge of $P = F/A$; making and recording observations with a syringe system; analysing and interpreting data (NGSS SEP: Analyzing and Interpreting Data); using mathematics and computational thinking to apply $P = F/A$ and $F_1/A_1 = F_2/A_2$; developing and using models (NGSS SEP) by drawing a hydraulic jack; constructing explanations from evidence; communication and collaboration in groups of four.
Attitudes	Curiosity about how a small push can lift a matatu; responsibility when handling equipment and sharing roles; respect for the ideas of others during discussion; appreciation of how Physics is applied by Kenyan mechanics such as Mzee Otieno.
Key Inquiry Question	How is pressure transmitted in fluids? (Also supports: How is pressure used in everyday life?)
Purpose in Storyline	Lessons 2-3 gave us $P = F/A$ and showed that, for the same force, a smaller area gives a larger pressure (the sharp knife: 8 MPa vs the blunt knife: 0.8 MPa, both from the same 40 N). Lesson 5 showed that pressure in liquids acts in all directions. In Lesson 8 learners see that pressure can be carried through an enclosed fluid and that area can be used to change the force. The same idea is at work in the knives: the force is the same and the area makes the difference. Here the pressure is the same (100 kPa) and the area makes the force different (50 N vs 1,000 N). This prepares Lesson 9, where hydraulics are consolidated with the knife phenomenon to complete the class model.
Safety Notes	Use syringes WITHOUT needles. Use only water (no needles, no sharp objects). Wipe up spills at once, as wet floors are slippery. Do not point syringe jets at faces or eyes. Do not push pistons with great force, because tubing can burst off. Demonstrate the real hydraulic jack only from a diagram or video; never stand under a lifted load. Students keep fingers clear of the plunger tips and tubing connections.

B. LESSON OVERVIEW

Mama Wanjiru's family garage mechanic, Mzee Otieno, lifts a heavy matatu with a hydraulic jack using only a 50 N push, yet the matatu needs about 1,000 N to be lifted. Learners connect this to the kitchen: with the knives, the same force gave different results because of area. Now the opposite happens: the same pressure (100 kPa) acts on different areas and gives different forces. In groups, learners first predict, then observe two linked syringes (small and large) filled with water, then use the data of the jack (5 cm² and 100 cm², 50 N, 100 kPa, 1,000 N) to explain Pascal's principle. They add a new question to the DQB and extend the class model to show that pressure is transmitted through a fluid and that $F = P \times A$ on each piston.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Pressure and Pascal's principle (part 1) - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Pascal's Principle (Flexbook) Source: CK-12  Search ARES for similar readings	(About 7 minutes) Learners see a drawing of Mzee Otieno's garage in Kisumu: a matatu on a hydraulic jack, with a small piston (5 cm ²) pushed by a 50 N effort and a large piston (100 cm ²) under the matatu. In groups of four, they answer on a Predict sheet: (1) 'Mzee Otieno pushes with only 50 N. Can the jack lift a load of 1,000 N? Yes or no, and why?' (2) 'If you push the small piston, what happens to the water in the closed tube? Draw arrows to show where the push goes.' (3) 'Link to the knives: Mama Wanjiru used the same 40 N. What stayed the same here, and what changed?' They draw their prediction and write one sentence beginning 'I think...'.	Display the drawing on the board. Say: 'Mzee Otieno lifts a whole matatu with just 50 N from his hand. Turn to your group: do you think this is possible? Why?' Give 10 seconds of WAIT TIME after asking, before taking any answers. Circulate and listen for ideas such as 'the water pushes up', 'the big piston has more area' or 'it is magic'. Cold-call 3 groups: 'Group 2, what did you draw for the arrows in the water?' Do NOT say whether the predictions are right. Say: 'Keep your prediction sheets; we will test them with real syringes.' Recall: 'Remember in Lesson 5 we found that liquid pressure acts in all directions. Keep that in mind.'	Prediction with drawn models (Predict-Observe-Explain): by drawing arrows before seeing evidence, learners reveal their ideas about how a push moves through water. Learners must commit to a prediction, so the later observation challenges or confirms their thinking. Linking back to the knives makes them compare the 'same force, different area' situation with the new situation.	Look for: (a) how many groups draw arrows only in the direction of the push, versus arrows in all directions (signals understanding of Lesson 5); (b) whether students mention area or 'bigger piston' in their reasoning; (c) whether any group says the force must be equal on both pistons (a misconception to note). Tally predictions on the board as Yes / No / Not sure. Record which groups mention area or pressure.
Observe Phase  VIDEO: Pressure and Pascal's principle (part 2) - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Piston Pressure (PLIX) Source: CK-12  Search ARES for similar readings	(About 12 minutes) In groups of four, learners use two water-filled syringes (no needles) joined by a short clear tube: a small syringe (e.g. 5 mL) and a large syringe (e.g. 20 mL). They carry out three tests. Test 1: Push the small plunger and watch what the large plunger does. Test 2: Push the large plunger and feel how much force is needed to move the small plunger. Test 3: Compare how far each plunger moves (use a ruler to measure the distance in cm). They also squeeze one syringe with a plugged tube and notice that the water pushes back on the finger. Then they study the data table from the real hydraulic jack: small piston area 5 cm ² (0.0005 m ²), large piston area 100 cm ² (0.01 m ²), effort 50 N, pressure 100 kPa, load 1,000 N. Roles: Pusher, Measurer, Recorder, Reporter.	Hand out the syringes and give safety reminders: 'Water only, no needles, point the tubes away from faces.' Say: 'Observe, do not explain yet. Write what you see and feel.' Ask probing questions while circulating: 'Which plunger moved farther? How can you tell?' 'Where did you feel the push needed to be bigger?' 'If you remove all the air bubbles, does it change anything?' Give 10-15 seconds WAIT TIME after each probing question. Check that students record measured distances (for example, the small plunger moves about 4 cm while the large plunger moves about 1 cm, depending on the syringes). Then show the hydraulic jack data table on the board. Ask: 'What do you notice about the pressure of 100 kPa? Is it the same at the small piston and the large piston?' Cold-call 3 students to read out the	Structured observation with a data table: learners separate what they see (plunger distances, effort felt) from what they think. Comparing the physical syringes with the jack data helps them see the same pattern at two scales. Recording numbers with units prepares them to find a mathematical relationship in the Explain Phase.	Look for: (a) students record that the small plunger moves farther than the large plunger; (b) students describe that pushing the small plunger needs less effort than pushing the large one for the same movement; (c) students include units (cm, N, Pa) in their tables; (d) students notice the water 'pushes back' in all directions. Check with a quick thumbs-up/thumbs-down: 'Did the force on the large plunger feel bigger than on the small one?' Note groups that confuse force with pressure.

		figures. Do not yet give the formula for the relationship.		
Explain Phase  VIDEO: Pressure and Pascal's principle (part 1) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Pascal's Principle (Flexbook) Source: CK-12  Search ARES for similar readings	(About 12 minutes) Learners use the jack data to make sense of what they observed. Step 1: Calculate the pressure on the small piston: $P = F / A = 50 \text{ N} / 0.0005 \text{ m}^2 = 100,000 \text{ Pa} = 100 \text{ kPa}$. Step 2: State that the enclosed water transmits this pressure equally in all directions, so the pressure on the large piston is also 100 kPa (Pascal's principle). Step 3: Find the force on the large piston: $F = P \times A = 100,000 \text{ Pa} \times 0.01 \text{ m}^2 = 1,000 \text{ N}$. Step 4: Show that $F_1/A_1 = F_2/A_2$: $50 \text{ N} / 0.0005 \text{ m}^2 = 1,000 \text{ N} / 0.01 \text{ m}^2 = 100,000 \text{ Pa}$. Step 5: Notice that the large piston has 20 times more area ($100 \text{ cm}^2 / 5 \text{ cm}^2 = 20$), so the load force is 20 times the effort ($1,000 \text{ N} / 50 \text{ N} = 20$). Step 6: Explain why the small piston moves 20 times farther than the large piston (the same volume of water moves). Step 7: Apply the idea to syringes (a small push on the plunger produces pressure that pushes the medicine out) and to matatu brakes (the foot pushes a small piston in the master cylinder; pressure travels through brake fluid to larger pistons at the wheels). Finally, they compare with the knives: for the knives the same force on different areas gave different pressures (8 MPa vs 0.8 MPa); for the jack the same pressure on different areas gives different forces (50 N vs 1,000 N).	Guide the steps on the board and have learners calculate in their books. Ask: 'Using $P = F/A$ from Lesson 2, what is the pressure on the small piston?' Give 10 seconds WAIT TIME, then cold-call 2 students to give the answer with the unit. Ask: 'The water is enclosed. The pressure is 100 kPa at the small piston. What do you think the pressure is at the large piston?' Wait 10-15 seconds, take 3 answers, then say: 'This is Pascal's principle: pressure applied to an enclosed fluid is transmitted equally in all directions.' Ask: 'If the pressure is the same but the area is 20 times bigger, what happens to the force?' Cold-call 3 students. Challenge: 'Does the jack give us free energy? Look at the distances: the small piston moves 20 times farther.' Ask a student to explain with the syringes' measured distances. Prompt for the link: 'In the kitchen, same force, different area gave different pressure. Here, same pressure, different area gives different force. What stays the same and what changes?' Address misconceptions: 'Pressure is the same on both pistons, but the forces are not.' Ask them to explain how a mechanic like Mzee Otieno lifts a matatu and how brakes in a matatu stop it using a small foot push.	Claim-Evidence-Reasoning with mathematical modelling: learners make the claim (a small force can lift a large load), use the evidence (syringe measurements, jack data) and the reasoning ($P = F/A$, pressure transmitted equally, $F_1/A_1 = F_2/A_2$). Comparing the jack to the knives (same force, different area vs same pressure, different area) builds a deeper understanding of the pressure-force-area relationship and prevents the idea that the jack creates energy.	Look for: (a) correct calculation of 100 kPa with the unit; (b) correct use of $F = P \times A$ to get 1,000 N; (c) a statement that pressure is equal on both pistons while forces differ; (d) a correct explanation that the small piston moves farther; (e) ability to link brakes or syringes to the principle. Use a 'Show me' check: learners hold up a card to say whether the pressure on the large piston is more than, equal to, or less than that on the small piston (the answer is equal). Note the learners who write that force is equal on both pistons, and revisit with them.
Driving Question Board (DQB) Creation  VIDEO:	(About 4 minutes) Learners return to the Driving Question Board with the main question 'Why does a sharp knife cut better than a blunt knife when using the same force?'	Say: 'Take 1 minute with your group and write one thing this lesson added to our answer. Remember, we are still trying to explain the knives.' Give 10	Driving Question Board as a living record: adding what we learned and what we still wonder makes learning visible, connects each lesson to the phenomenon, and	Look for: (a) notes that mention both 'pressure' and 'area' or 'force'; (b) at least one note that links the jack to $P = F/A$; (c) new questions that show curiosity (siphon,

Pressure and Pascal's principle (part 2) Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Piston Pressure (PLIX) Source: CK-12 Search ARES for similar readings	Groups write one new idea on a sticky note under 'What we learned' (for example, 'Pressure is transmitted equally through a closed fluid, and a bigger area gives a bigger force at the same pressure') and one remaining question under 'What we still wonder' (for example, 'How does a siphon move water without a pump?' or 'Can we use the area idea to sharpen a panga?'). They place sticky notes next to related questions such as 'How is pressure transmitted in fluids?' and 'How is pressure used in everyday life?'.	seconds of WAIT TIME after the prompt. Invite 3 groups to place their notes and read them aloud. Ask: 'How does the jack help us think about the knives? What is the same? What is different?' Group together questions on the DQB and point out the ones that Lesson 9 will address (the siphon, dams and consolidation). Praise accurate use of units.	gives learners ownership of the next steps. Placing notes by question makes learners connect evidence with questions.	brakes, dams); (d) notes that wrongly say the jack 'makes energy' - flag these for Lesson 9. Count how many groups contribute at least one note.
Model Building Phase VIDEO: Pressure and Pascal's principle (part 1) - dubbed(KY) Source: Khan Academy (Kiswahili) Search ARES for similar videos EXERCISE: Pressure in Fluids Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	(About 5 minutes) Learners revise their individual models of the knife phenomenon in their notebooks. They add a small 'pressure in fluids' box showing the hydraulic jack: two pistons of different sizes joined by an enclosed fluid, with labels 'F1 = 50 N, A1 = 0.0005 m², P = 100 kPa' and 'F2 = 1,000 N, A2 = 0.01 m², P = 100 kPa'. Arrows in all directions show the pressure in the fluid. Next to the box they write the rule 'P = F / A' and 'F1/A1 = F2/A2'. On the knife part of the model, they write: 'Same force 40 N, smaller area gives bigger pressure (8 MPa vs 0.8 MPa)'. They add one sentence: 'In the jack, the same pressure acts on a bigger area, so the force is bigger.' Learners then complete the Summary Table.	Say: 'Revise your model. Add the jack and show how it is the same idea as the knives, in a different situation. Use arrows, labels and units.' Move around and check that models include units, pressure arrows in all directions and the equal pressure of 100 kPa on both pistons. Give 10-15 seconds WAIT TIME before asking: 'What part of your model explains why the matatu lifts?' Cold-call 2 learners to present their model to the class. Prompt: 'What would you change if the large piston were 200 cm²?' (Answer: the load would be 2,000 N for the same 50 N effort, because F = 100,000 Pa x 0.02 m².) Remind them that Lesson 9 will finalise the model.	Cumulative model revision: learners add the new idea to their existing model instead of starting again. Showing the knife and the jack side by side in the same model helps them see that $P = F/A$ explains both. Using labelled arrows and units turns the model into a tool for predicting.	Look for: (a) a model with two pistons of different sizes and an enclosed fluid; (b) the same pressure (100 kPa) shown on both pistons; (c) correct values and units (50 N, 1,000 N, 5 cm² or 0.0005 m², 100 cm² or 0.01 m²); (d) an explicit link to the knives ('same force, different area'); (e) the rule $F1/A1 = F2/A2$. Use a quick exit question: 'Why does the jack lift 1,000 N with only 50 N?' Look for the answer: 'The pressure is the same on both pistons (100 kPa), and the large piston has 20 times the area, so the force is 20 times larger.'

D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners separate pressure (equal on both pistons, 100 kPa) from force (50 N vs 1,000 N)? Which groups still said the forces are equal? (2) Could learners link the jack back to the knives ('same force, different area' vs 'same pressure, different area')? (3) Were the syringe measurements clear enough, or do the air bubbles need more attention next time? (4) Did learners misunderstand the jack as 'free energy'? Note who needs the distance argument again in Lesson 9.

(5) Which learners used units correctly, and who needs more practice converting cm^2 to m^2 ($1 \text{ cm}^2 = 0.0001 \text{ m}^2$)? (6) Did cold-calling and 10-15 seconds of WAIT TIME give quieter learners a chance to speak? Prepare for Lesson 9 by choosing sticky notes about the siphon and dams, and plan how to bring the knives (8 MPa vs 0.8 MPa vs 2 MPa), the dam, the jack and atmospheric pressure together.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that when we pushed the small syringe plunger, the water in the closed tube moved the large plunger. The small plunger moved farther than the large one. The jack data showed a 50 N effort on a 5 cm^2 piston and a 1,000 N load on a 100 cm^2 piston, both with a pressure of 100 kPa.
What did I learn?	We learned that pressure applied to an enclosed fluid is transmitted equally in all directions (Pascal's principle). Since $F_1/A_1 = F_2/A_2$, the large piston, with 20 times the area, gives 20 times the force (1,000 N from 50 N). This is used in syringes, hydraulic jacks and brakes.
How does this explain the phenomenon?	We can explain that the knives and the jack both depend on $P = F/A$. For the knives the force is the same (40 N) and the area is different, so the sharp knife gives 10 times more pressure (8 MPa vs 0.8 MPa). For the jack the pressure is the same (100 kPa) and the area is different, so the force is different (50 N vs 1,000 N). Area is the key in both cases.

LESSON 9 (80 minutes (double lesson)): Applications and Consolidation: Siphon, Dams, Hydraulics and the Final Explanation of the Knife Phenomenon

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners consolidate Lessons 1-8 by applying pressure ideas to a siphon, a dam and a hydraulic jack, then finalise the class model and use it to give a complete evidence-based explanation of why Mama Wanjiru's sharp knife cuts while the blunt knife only squashes, with the same 40 N force.
Knowledge	Learners will be able to: (1) explain that pressure is force per unit area, $P = F/A$, and that for the same force a smaller area gives a greater pressure; (2) use the knife data (40 N; $5 \times 10^{-6} \text{ m}^2$ giving 8 MPa; $5 \times 10^{-5} \text{ m}^2$ giving 0.8 MPa; food gives way at about 2 MPa) to explain the phenomenon; (3) explain why a dam wall is thicker at the bottom (300 kPa at 30 m compared with 100 kPa at 10 m); (4) explain that a siphon works because of atmospheric pressure (about 101 kPa at sea level) and the difference in depth of the liquid; (5) explain that a hydraulic jack uses Pascal's principle so that a 50 N effort gives a 1,000 N load ($F_1/A_1 = F_2/A_2$). No new formula is taught.
Skills	Developing and using models (revising the class model); Constructing explanations (claim-evidence-reasoning); Analysing and interpreting data (comparing pressures with the 2 MPa threshold); Using mathematics (substituting in $P = F/A$, $P = \rho g h$ and $F_1/A_1 = F_2/A_2$ with units); Engaging in argument from evidence; Communication and collaboration.
Attitudes	Appreciate how the physics of pressure explains everyday Kenyan situations (kitchen, dam, garage); show curiosity and persistence when linking ideas; respect others' ideas in group discussion; take responsibility for accurate recording and care when handling knives (teacher-only) and water.
Key Inquiry Question	Why does a sharp knife cut better than a blunt knife when using the same force? (Consolidating all seven KICD key inquiry questions: What is pressure? How does area affect pressure? What factors affect pressure in fluids? Why does pressure increase with depth? What is atmospheric pressure? How is pressure transmitted in fluids? How is pressure used in everyday life?)
Purpose in Storyline	Final lesson of the storyline. Lessons 1-8 built the pieces: the knife observations (L1), $P = F/A$ (L2), area and the knife calculations (L3), liquid pressure and depth (L4-6), atmospheric pressure (L7) and Pascal's principle (L8). Lesson 9 brings them together in three applications, completes the Driving Question Board, compares the Lesson 1 model with the final model, and delivers the Final Explanation of the knife phenomenon.
Safety Notes	Mama Wanjiru's knives are NOT handled by learners; the teacher demonstrates, or learners use a plastic knife and soft tomato or clay. Wipe up spilt water immediately to prevent slipping. Siphon tube: use clean water and do not suck the tube by mouth; fill the tube by submerging it. Keep the hydraulic syringe set-up on a tray. Handle the glass tumbler (if used) over a basin.

B. LESSON OVERVIEW

This final lesson is a synthesis lesson with no new formula. Learners start by predicting how the knife explanation, a siphon, a dam and a hydraulic jack are linked, then observe evidence in three short stations: (A) a siphon moving water between two containers (atmospheric pressure and depth difference), (B) a dam diagram with pressures of 100 kPa at 10 m and 300 kPa at 30 m, and (C) a two-syringe hydraulic model matched to Mzee Otieno's jack (50 N giving 1,000 N). In the Explain phase, learners write a claim-evidence-reasoning Final Explanation of the knife phenomenon using 40 N, $5 \times 10^{-6} \text{ m}^2$ and $5 \times 10^{-5} \text{ m}^2$, 8 MPa versus 0.8 MPa and the 2 MPa threshold, stating 'same force, different area'. They then complete the Driving Question Board, mark which questions have been answered and by which lesson, and finally compare the Lesson 1 initial model with the final class model, noting what changed and why.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Pressure and Pascal's principle (part 1) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Pascal's Principle (Flexbook) Source: CK-12  Search ARES for similar readings	(10 min) Learners look at the Lesson 1 initial model and the DQB on the wall. Individually, for 2 minutes, they write on a 'Predict Card': (a) one sentence completing 'The sharp knife cuts better because...' and (b) a prediction answering 'Which ONE idea from our pressure lessons will also help explain a dam wall, a siphon and Mzee Otieno's jack that lifts a matatu with only 50 N?' They pair up, share, and add a confidence rating (1-3).	Display the DQB and L1 model. Say: 'Today is our last lesson. Take two minutes of silent thinking and write your best answer to the question: why does Mama Wanjiru's sharp knife cut when the blunt one only squashes, with the same force?' WAIT TIME: 10-15 seconds after asking: 'What do a knife, a dam, a siphon and a jack have in common?' before taking any answer. Cold-call 4 learners (two who answered with 'force' and two who mention 'area' or 'pressure') and record their ideas on the board in two columns. Do not correct yet; say: 'Hold on to your prediction; we will test it with evidence.'	Predict-and-Commit with Confidence Rating: writing a prediction and a confidence level before new evidence makes learners' current thinking visible, so that they can later notice how their ideas changed and why. Linking the earlier model to the prediction activates prior knowledge from Lessons 1-8.	Look for: (1) at least 6 of 10 predictions mention force AND area, or the word 'pressure'; (2) learners who say 'the sharp knife has more force' (misconception) are noted for follow-up; (3) learners link at least one other application (dam, siphon, jack) to pressure. Record counts of correct, partially correct and misconception predictions on a tally sheet.
Observe Phase  VIDEO: Pressure and Pascal's principle (part 1) - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Pressure in Fluids (Flexbook) Source: CK-12  Search ARES for similar readings	(25 min) Groups of 4 rotate through three 7-minute stations, recording observations and data in a table (Observed / Data / Link to pressure). Station A, Siphon: a clear tube is filled with water by submerging it, then one end is placed in a higher bucket and the other in a lower basin; learners observe the flow, then raise the outlet to the same level as the water surface and observe that flow stops. They note that the liquid depth difference and atmospheric pressure (about 101 kPa at sea level) push the water through the tube; the air pushes down on the water surface in the bucket. Station B, Dam: a dam diagram (Masinga-type dam, 30 m deep) with a wall thicker at the bottom; learners read pressure at the 10 m level (100 kPa) and at the base 30 m (300 kPa) using $P =$	Set up the stations before the lesson and give each group a data sheet. Say: 'At every station ask yourselves: where is the force, where is the area, and what is the pressure?' Circulate and ask each group at least one question: 'At Station A, what happens when the outlet is raised to the water level? What does that tell you about the depth difference?' 'At Station B, why is the wall thicker at the bottom? Use the numbers 100 kPa and 300 kPa.' 'At Station C, the effort is only 50 N; where does the extra force come from? Check the area ratio of 20.' Use WAIT TIME of 10-15 seconds after each question before prompting. Stop the rotation with a clear signal at 7 minutes. For the whole-class demonstration (teacher only handles the knives), ask: 'Both pushes are the same force, 40 N	Station Rotation with Data Table (Analysing and Interpreting Data): collecting observations at three different applications and asking the same three questions (force? area? pressure?) at each station helps learners see that one idea, pressure = force/area, and its extensions explain many different situations, including the knife phenomenon.	Check data sheets for: (1) Station A: learners write that the water flows from the higher to the lower level and stops when the levels are equal, and mention air/atmospheric pressure; (2) Station B: correct values of 100 kPa and 300 kPa with units; (3) Station C: correct matching of 50 N giving 1,000 N, area ratio 20, and recognition that the small piston moves farther. Use a checklist while circulating; note groups that omit units or confuse force with pressure.

	$\rho g h = 1,000 \times 10 \times 30 = 300,000 \text{ Pa}$, and note the base has 3 times the pressure of the 10 m level. Station C, Hydraulic jack model: two syringes of different sizes joined by a tube filled with water; pushing the small one moves the large one with greater force. They match this to Mzee Otieno's jack: small piston $5 \text{ cm}^2 = 0.0005 \text{ m}^2$, large piston $100 \text{ cm}^2 = 0.01 \text{ m}^2$, effort 50 N gives 100 kPa, load 1,000 N, and the small piston moves 20 times farther than the large one. A fourth, short whole-class observation: the teacher cuts a soft tomato with a sharp knife and a blunt knife using the same gentle push, while learners watch from a distance.	in our model. What do you see?' and cold-call 3 learners to describe the sharp and blunt results.		
Explain Phase  VIDEO: Pressure and Pascal's principle (part 2) - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Atmospheric Pressure (Flexbook) Source: CK-12  Search ARES for similar readings	(20 min) Part 1 (8 min): In groups, learners explain each application using a 'Because...' chain. Siphon: air pushes down on the water surface (atmospheric pressure about 101 kPa), the water in the tube is pushed through, and flow continues while there is a depth difference. Dam: pressure increases with depth ($P = \rho g h$), so the base at 30 m has 300 kPa and the wall is built thicker there. Jack: pressure applied to an enclosed fluid is transmitted equally (Pascal's principle), so $F_1/A_1 = F_2/A_2$ gives 50 N becoming 1,000 N with a 20 times bigger area. Part 2 (12 min): Individually, learners write the FINAL EXPLANATION of the knife phenomenon in the Claim-Evidence-Reasoning format. Claim: the sharp knife cuts because it exerts a much greater pressure on the food, since the area is smaller with the same force. Evidence: Mama Wanjiru	Say: 'You now have three applications and one big idea. Explain each one in a chain using the word because.' Model one chain on the board for the dam. For the Final Explanation, display the CER frame and say: 'Your explanation must include the words same force, different area, and all four numbers: 40 N, $5 \times 10^{-6} \text{ m}^2$, $5 \times 10^{-5} \text{ m}^2$, 8 MPa and 0.8 MPa, and the 2 MPa threshold.' WAIT TIME: 10-15 seconds after the question: 'If Mama Wanjiru pushed harder with the blunt knife, say 400 N, what would happen to the pressure? What else could she do instead?' Cold-call 3 learners; guide to: 'A greater force would increase pressure, but sharpening reduces the area, which is easier and gives the same effect.' Do not introduce new formulas. Then ask 3 learners to read their claim and 2 to read their reasoning; ask the class to give thumbs up/down on whether	Claim-Evidence-Reasoning (Constructing Explanations): requiring a claim supported by numerical evidence and linked by the scientific principle $P = F/A$ forces learners to connect data to theory rather than recall facts. Using 'because' chains for the applications shows the same pressure ideas at work in different contexts, building transferable understanding.	Use a 4-point rubric on each Final Explanation: (1) correct claim about pressure and area (1 point); (2) all key numbers with units, i.e. 40 N, 8 MPa, 0.8 MPa, 2 MPa (1 point); (3) uses $P = F/A$ and 'same force, different area' (1 point); (4) compares each pressure with 2 MPa to explain cut vs squash (1 point). Target: at least 75% of learners score 3 or 4. Note any learner who writes 'more force' or who says 'stress' instead of 'pressure' and conference with them.

	pushes with 40 N on both knives; sharp edge area = $0.05 \text{ m} \times 0.0001 \text{ m} = 5 \times 10^{-6} \text{ m}^2$, pressure = $40/0.000005 = 8,000,000 \text{ Pa} = 8 \text{ MPa}$; blunt edge area = $0.05 \text{ m} \times 0.001 \text{ m} = 5 \times 10^{-5} \text{ m}^2$, pressure = $40/0.00005 = 800,000 \text{ Pa} = 0.8 \text{ MPa}$; tomato or nyama choma gives way at about 2 MPa. Reasoning: $P = F/A$; same force, different area; the sharp edge has 10 times less area so 10 times more pressure; 8 MPa is greater than 2 MPa so the food is cut, 0.8 MPa is less than 2 MPa so the food is only squashed; the difference between the knives is the area, not the force.	the reasoning uses area, not force, as the difference. Correct the misconception 'the sharp knife has more force' by pointing to 40 N on both.		
Driving Question Board (DQB) Creation  VIDEO: Pressure and Pascal's principle (part 2) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Atmospheric Pressure Practice Source: CK-12  Search ARES for similar readings	(10 min) Learners return to the class DQB. In groups, they take the sticky notes of questions asked during Lessons 1-8 and sort them into 'Answered', 'Partly answered' and 'Still wondering'. For each answered question, a learner writes the answer on the note with the evidence and the lesson number (for example, 'Why does the sharp knife cut? Same force 40 N, smaller area, 8 MPa above 2 MPa, L3'). They match the seven KICD key inquiry questions to the evidence: pressure (L2), area (L3), liquids and depth (L4-6), atmospheric pressure (L7), transmission of pressure (L8), everyday use (L9: siphon, dam, jack). Finally each learner adds one NEW question for further inquiry (for example, 'Why do our ears pop when travelling from Mombasa to Nairobi?' or 'How do car brakes use Pascal's principle?').	Say: 'Our board started with wonderings in Lesson 1. Which ones can we now answer with evidence?' Give each group 4 minutes to sort and 3 minutes to write answers. WAIT TIME: 10-15 seconds after asking: 'Which question took the most lessons to answer and why?' before taking responses. Cold-call 4 groups to read one answered question with its evidence and lesson. Point out any question that is still unanswered and invite learners to think of how they could investigate it. Ask: 'Which DQB question is the most important for explaining the knife?' and guide learners to the area question.	Driving Question Board Review (Questioning and Metacognition): tracing every question to the evidence that answered it shows learners the path of their own thinking, and keeps the whole unit anchored in the knife phenomenon. Asking for new questions develops curiosity and shows science as ongoing inquiry.	Observe whether: (1) at least 80% of the original questions are sorted correctly and answered with specific evidence (numbers with units); (2) each group links answers to the correct lesson; (3) each learner writes at least one new, testable question. Collect the new questions as an exit slip.
Model Building	(15 min) Part 1 (5 min): Learners take out or view the Lesson 1	Say: 'A good model shows what you cannot see: the force, the area	Model Revision and Comparison (Developing and Using Models):	Check models against the criteria: (1) same 40 N force on both

Phase  VIDEO: Pressure and Pascal's principle (part 1) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Pascal's Principle (Flexbook) Source: CK-12  Search ARES for similar readings	initial model (their first drawing and ideas about the knives). Part 2 (8 min): In groups, they create the FINAL CLASS MODEL on large paper: a drawing of the sharp and blunt knife edges pressing into a tomato, with arrows showing the same force of 40 N, the contact areas labelled $5 \times 10^{-6} \text{ m}^2$ and $5 \times 10^{-5} \text{ m}^2$, the pressures 8 MPa and 0.8 MPa, a threshold line at 2 MPa, and the words $P = F/A$. Around it they add three small panels showing links: dam (300 kPa at 30 m compared with 100 kPa at 10 m), siphon (atmospheric pressure about 101 kPa and depth difference) and hydraulic jack (50 N giving 1,000 N). Part 3 (2 min): Gallery walk; each learner completes a 'Model Comparison' strip: 'In Lesson 1 I thought... Now I know... because (evidence)...'. The class then agrees on one consensus model that is displayed beside the Lesson 1 model.	and the pressure at the edge of the knife.' Give a checklist: force arrows labelled, areas labelled, pressures with units, threshold, and the panels. WAIT TIME: 10-15 seconds after asking: 'What is missing from our Lesson 1 model that we can now add?' Cold-call 3 learners; expect 'area', 'pressure', 'the 2 MPa threshold'. During the gallery walk, ask each group to defend one choice: 'Why did you draw the sharp edge so thin?' Conduct a final whole-class discussion: ask 3 learners to read their comparison strip and ask the class: 'Which change in our thinking was the biggest?' Close by reading the class's Final Explanation aloud: 'Same force, different area, so different pressure.'	placing the Lesson 1 model next to the final model makes the growth in understanding visible, helping learners see how evidence changed their thinking from 'force' to 'pressure = force/area'. Adding application panels shows the model's power to explain new situations.	knives; (2) correct areas and pressures with units; (3) 2 MPa threshold used to explain cut vs squash; (4) $P = F/A$ stated; (5) at least two correct application panels. Check that the comparison strip names a specific change and the evidence for it. Success: at least 80% of groups meet 4 of 5 criteria. Record misconceptions still present for follow-up.
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D. TEACHER REFLECTION

Before the lesson: Have I set up the three stations safely and tested the siphon and syringe jack? Have I prepared the 2 MPa threshold display and the CER frame? During the lesson: Did learners consistently say 'same force, different area' and use the numbers 40 N, 8 MPa, 0.8 MPa and 2 MPa with units? How many still believed the sharp knife has more force, and what evidence changed their minds? Which station caused the most difficulty (siphon, dam or jack) and why? After the lesson: Compare the L1 and final models for a sample of learners; which misconceptions remain (for example, confusing force with pressure, or saying 'stress')? Which DQB questions are still unanswered, and how can I use them in the next unit? What would I change to help learners with weaker mathematics substitute correctly into $P = F/A$?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that Mama Wanjiru's sharp knife (edge 0.1 mm) cuts a tomato and nyama choma while the blunt knife (edge 1 mm) only squashes them, though the force is the same 40 N. We also observed a siphon moving water from a higher to a lower level, a dam wall that is thicker at the bottom, and a hydraulic model in which a small effort of 50 N gave a load of 1,000 N.
What did I learn?	We learned that pressure is force per unit area, $P = F/A$, measured in pascals (Pa). For the same force, a smaller area gives a greater pressure. Liquid pressure increases with depth (300 kPa at 30 m in a dam compared with 100 kPa at 10 m); atmospheric

	pressure (about 101 kPa at sea level) helps a siphon work; and pressure in an enclosed fluid is transmitted equally, so $F_1/A_1 = F_2/A_2$ lets a jack lift a matatu.
How does this explain the phenomenon?	Final Explanation: Mama Wanjiru pushes with the same force of 40 N on both knives. The sharp edge has an area of $5 \times 10^{-6} \text{ m}^2$ and the blunt edge $5 \times 10^{-5} \text{ m}^2$, so the sharp knife gives a pressure of 8 MPa and the blunt knife 0.8 MPa, which is 10 times less. Tomato and meat give way at about 2 MPa, so 8 MPa cuts and 0.8 MPa only squashes. Same force, different area, therefore different pressure; the difference between the knives is the area, not the force.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.